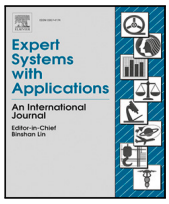
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A cost-minimized two-stage three-way dynamic consensus mechanism for social network-large scale group decision-making: Utilizing K-nearest neighbors for incomplete fuzzy preference relations

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ABSTRACT

In the era of big data, large scale group decision-making (LSGDM) with social networks (SNs) (namely, SN-LSGDM) has become a hot topic in the field of decision science. Faced with the explosive growth of information, decision-makers (DMs) face immense challenges in processing and integrating vast amounts of data, often finding it difficult to fully comprehend all the information, leading to potentially incomplete expressions of their fuzzy preference relations (FPRs). This limitation in information processing not only affects the quality of decision-making but also increases the difficulty and cost of reaching a consensus. To overcome these challenges and enhance the efficiency and accuracy of decision-making, this paper designs a consensus model that minimizes adjustment costs in light of a dynamic trust network. Firstly, we introduce a measurement method based on K-nearest neighbor (KNN) information, which comprehensively considers the trust level of DMs and the similarity of preference relations, effectively filling in missing preference information and improving the completeness and accuracy of decision-making. In addition, an improved k-means clustering algorithm is adopted, which takes into account the mutual influences between DMs and the cost of unit adjustment. On this basis, a two-stage minimum adjustment cost consensus reaching mechanism based on three-way decision (TWD) is proposed, using comprehensive adjustment priority as the criterion for division, to achieve feedback adjustment at the individual and subgroup levels, ensuring the coordination and consistency of the decision-making plan. At the same time, an optimization model is introduced to achieve cost minimization. Through detailed case studies and comparative analysis, the feasibility and superiority of this method in practical applications have been demonstrated.

1. Introduction

In this section, we recall background, applications, research progress and limitations, motivations and innovations of large scale group decision-making (LSGDM).

1.1. Background

In today's digital era, the significance of group decision-making (GDM) is increasingly recognized. As society's reliance on data and quantitative modeling deepens, the traditional decision-making process led by human experts is gradually being marginalized (Kovalchuk, Krotov, Smirnov, Nasonov, & Yakovlev, 2018; Long, 2018). Despite this, in certain fields, the involvement of experts remains crucial for ensuring the quality and relevance of decisions (Gilbert & O'Leary, 1975; Palomares, Rodríguez, & Martínez, 2013). However, with technological advancements, particularly in the fields of computer science, artificial intelligence, and machine learning, a revolution in the field of decision-making is evident (Avramov, Cheng, & Metzker, 2023). For instance, Protic and Stankovic (2023) put forth an XOR-based detector for anomaly detection in computer network traffic, demonstrating the potential of machine learning techniques in enhancing decision accuracy. These technologies not only provide powerful tools for processing and analyzing big data but also open new avenues for solving complex problems.

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1.2. Applications of LSGDM

Large scale group decision-making (LSGDM), as an emerging decision-making model (Garcia-Zamora, Labella, Ding, Rodriguez, & Martinez, 2022; Meng, Tang, & An, 2023), involves a large number of experts or decision-makers (DMs) working together to identify the best choice from a multitude of options (Xu, Du, & Chen, 2015). This model has been widely applied in key areas such as:

(1) Supplier selection (Chen, Zhu, Wang, & Tsang, 2023): A novel approach for generating group opinions by aggregating probability distribution functions to address complex decision analysis tasks, with a focus on fairness among experts.
(2) Healthcare resource allocation (Mardani, Zavadskas, Fujita, & Koppen, 2022): The application of big data methods for resource allocation under uncertainty, integrating expert opinions through clustering and consensus techniques.
(3) Electronic markets (Karthik & Ganapathy, 2021): A recommendation system based on fuzzy logic that analyzes customer reviews to offer personalized product suggestions. Social media (Sueur, Deneubourg, & Petit, 2012): Simulations of various social network structures to predict consensus patterns, demonstrating the influence of network structures.
(4) Earthquake shelter selection (Xu, Wen, & Zhang, 2018): Utilizing clustering algorithms and GIS analysis to streamline decision-making and assess shelter feasibility in emergency contexts.
(5) Water resource management (Gou, Xu, & Herrera, 2018; Zhang, Xiao, Palomares, Liang and Dong, 2020): Designing a framework optimized for linguistic distribution to facilitate large group decision-making in wastewater treatment technology selection. Developing a consensus-building framework for water resource management policy implementation, using hesitant fuzzy linguistic preferences and information entropy for weight determination.

Furthermore, advancements in communication technology have facilitated interactions among DMs, making the study of interactions within social networks (SNs) among DMs an intriguing new research topic.

1.3. Research progress and limitations

In this subsection, we introduce research progress of LSGDM and point out its limitations.

1.3.1. Social networks

In recent years, LSGDM with SNs (namely, SN-LSGDM) have attracted much attention to many scholars (Liu, Zhang, Wang, & Wang, 2022; Lu, Xu, Herrera-Viedma, & Han, 2021). This issue involves multiple DMs with different opinions, interacting through SNs to reach a consensus (Du, Luo, Lin and Yu, 2020; Li, Kou, Li, & Peng, 2022). Studies have shown that social relationships play an important role in GDM, such as affecting information transmission, DMs' weight distributions, and the consensus reaching process (CRP) (Wu, Liu, Qin, & Herrera, 2021). To address the challenges of SN-LSGDM, scholars have proposed key steps such as clustering, consensus and selection. Clustering methods group DMs based on similarity measurement to address the scalability issue of decision-making. Moreover, the consensus process uses social trust relationships to reduce opinion differences and promote consensus. Social trust network (STN) techniques such as community detection and network partitioning are widely used in this issue, effectively utilizing the social relationships between DMs. For example, Lu, Xu, Huang, Wei, and Herrera-Viedma (2022) explored an integrated approach involving STN clustering and consensus-based distrust behavior management, along with the development of corresponding consensus management processes and strategies for distrust behavior identification and modification. Gong, Ma, Lei, and Goh (2024) constructed a dynamic trust network by integrating the opinion similarity and trust relationships among DMs, and designed a mechanism for the dynamic updating of the trust network and a two-stage feedback adjustment process. In summary, SNs bring new opportunities and challenges to SN-LSGDM research, which need further exploration and resolution.

1.3.2. Addressing incomplete information

One of the most prevalent challenges is the issue of data incompleteness, a topic that has been extensively discussed in the field of decision science and numerous other research disciplines. In practice, DMs often face incomplete or uncertain information, which hampers the efficacy of conventional models. Ketsarapong, Punyangarm, Phusavat, and Lin (2012) addressed this by creating a fuzzy logic-based inventory system that leverages expert experience to make informed decisions without relying on exact data. Yan, Zhao, Xie, and Precup (2024) proposed an online semi-supervised ensemble fuzzy system called an OSSEFS, which is designed to effectively handle data stream learning problems with missing values and unlabeled data.

Currently, the existing models for incomplete fuzzy preference relations (IFPRs) largely rely on consistency methods and statistical learning approaches to estimate missing values. Consistency methods are categorized into two types: additive consistency and multiplicative consistency (Tanino, 1984). Chen and Xu (2019) introduced a method for incomplete intuitionistic FPRs in GDM, focusing on additive consistency and algorithms for correcting inconsistencies. Zhang, Wang, and Tian (2016) put forth an estimation method for missing values in hesitant FPRs based on an additive consistency. Xie, Ma, Xu, Fu, and Yang (2022) developed the concept of generalized intuitionistic FPRs, providing standards for consistency and a threshold-free method to preserve original preferences in incomplete cases. Song and Li (2019) established a mathematical optimization model to achieve multiplicative consistent FPRs. Xu, Li and Wen (2018) designed a normalization and missing element estimation method for incomplete hesitant FPRs based on multiplicative consistency. However, these consistency-based methods may lead to the loss of original preference information when adjusting known preference values to meet consistency conditions. Moreover, the assumptions of additive and multiplicative consistency may not be applicable to actual preference relations in some cases, limiting the applicability of these methods.

Additionally, many scholars have explored statistical learning methods. For instance, Chu, Wang, Liu, and Liu (2020) revealed a repair method for incomplete FPRs based on community division. Wu, Tian, Lan, and Li (2023) introduced an estimation method using sequential KNN interpolation and attribute-weighted k-means clustering for incomplete and hesitant assessments, enhancing the representativeness of collective evaluations. However, this method assumes that the importance of each neighbor in K-nearest neighbors is consistent, which may not reflect the reality. Both methods overlook the reference risk that distrust and disagreement may pose. As the number of DMs involved in decision-making increases and social relationships become more complex, trust relationships have become a critical factor influencing the opinions of DMs. Lu et al. (2022) addressed the issues of incomplete hesitant FPRs and distrust in SN-LSGDM by employing grey clustering and distrust behavior correction mechanisms, thereby improving the estimation of missing values. However, this method's initial step of clustering DMs with missing information may affect the accuracy of the clustering results. In response to the aforementioned issues, this paper focuses on statistical learning methods and discusses how to effectively integrate STN into the process of estimating missing information. The complex trust and preference relationships within STNs offer a new perspective for estimating missing data, providing new directions for future research.

1.3.3. Consensus reaching process

The CRP is a pivotal procedure in managing SN-LSGDM events and aims at achieving highly consistent decision outcomes. However, the pursuit of absolute consensus can often lead to unnecessary expenditure of costs and resources. In response, the concept of soft consensus has been increasingly integrated into various CRP models (Herrera-Viedma, Cabrerizo, Kacprzyk, & Pedrycz, 2014; Xu, Du, Chen, & Cai, 2019). This approach advocates for more lenient measures of consensus, better reflecting all possible partial agreements and guiding the consensus process towards a high degree of unanimity among individuals. The existing CRP consensus models encompass a range of areas, including those that aimed to minimize costs or adjustments (Ben-Arieh, Easton, & Evans, 2009; Zhang, Zhao et al., 2020), those that applied in LSGDM scenarios (Du, Yu and Xu, 2020; Liu, Xu, Montes, & Herrera, 2019), those that conducted within SNs (Du, Yu, Luo, & Lin, 2021; Liu et al., 2019), and those that took into account the behaviors of DMs (Gou, Xu, Liao, & Herrera, 2021; Sun et al., 2023). The consensus achievement process can be categorized into one-stage and two-stage methods, each tailored to the specific requirements and dynamics of the decision-making environment. This evolution in CRP models acknowledges the practicality and flexibility needed in real-world decision-making, where a perfect consensus is not always feasible or desirable. By embracing soft consensus, these models strive for a more realistic and economically viable path to collective agreement, ultimately enhancing the efficiency and adaptability of the decision-making process in diverse and complex scenarios. In the process of consensus achievement, it is generally divided into one-stage and two-stage methods. Lu et al. (2022), based on grey clustering, managed and modified the distrustful behaviors of DMs to reach the final consensus. Wu, Liu, Qin, and Herrera (2022) introduced the concept of unit adjustment cost (UAC), solving the optimization model to achieve consensus with minimized adjustment costs. Although these methods aim to achieve overall consensus, they do not fully consider whether consensus has been reached within the subgroups obtained by clustering, which may lead to a decrease in decision quality and an increase in conflict risks. The two-stage method first reaches a consensus within the subgroups and then involves these subgroups as DMs in the overall consensus process. Gong et al. (2024) established a dynamic updating method for the trust network to determine the real-time weights of DMs and subgroups, and designed a two-stage feedback adjustment mechanism for individuals and subgroups. Shen, Ma, and Zhan (2023) closely integrated the clustering process with the consensus mechanism, proposing a two-stage adaptive consensus achievement model. This paper also adopts a two-stage consensus while ensuring decision quality and minimizing adjustment costs.

1.3.4. Three-way decision

Three-way decision (TWD) framework, introduced by Yao (2010), is a cognitive strategy adept at tackling decisions fraught with uncertainty and risk. This framework dissects the decision-making space into three distinct regions, each equips with tailored strategies. Scholars have increasingly integrated this approach with GDM, enriching its applicability. Notably, Ma, Lei, and Sun (2020) have advanced a three-way GDM model leveraging multi-granularity hesitant fuzzy decision theory across divergent regions. Meanwhile, Liang, Liu, and Kobina (2016) have contributed a rough set theory-based method for GDM within this three-way paradigm. However, the consensus in TWD has seldom considered the impact of STNs, a gap this paper aims to bridge, while also extending the discourse on the consensus in GDM traditionally confined to single-layer STNs.

1.3.5. Limitations

In practice, there are many practical challenges to implementing LSGDM. First, the complexity of information collection is the key issue. Especially when stakeholders are distributed in different sectors or regions, information sharing becomes difficult. Incomplete or asymmetric information often leads to decision-making bias, increasing adjustment costs and reducing consensus efficiency. Second, it is difficult to coordinate the opinions of all parties. Stakeholders have obvious differences in goals and priorities, especially in cross-departmental or cross-organizational projects, and these differences exacerbate conflicts in decision-making. Finally, industrial applications require high real-time decision-making, but the existing methods are difficult to cope with dynamic and complex information environments, which affect the timeliness and accuracy of decision-making.

In addition to the above challenges in practical applications, the existing LSGDM methods also have limitations in dealing with these complexities. In particular, when dealing with complex decision-making problems such as incomplete information and dynamic trust relationships, the existing methods are still insufficient and further in-depth research is urgently needed. Specifically, these limitations are mainly manifested in the following aspects:

(1) The limitation of dealing with incomplete information: Traditional decision-making models are often designed based on the completeness and certainty of preference information, while in actual applications, the information submitted by DMs is often incomplete or uncertain. This disconnection between assumption and reality limits the effectiveness of the model in the real world.

(2) The static nature of trust relationships: The existing methods tend to adopt static trust distribution, failing to capture and reflect the dynamic changes in the trust relationships between DMs, which affects the flexibility and adaptability of the decision-making process.

(3) The limitation of clustering algorithms: When dealing with individuals with different UAC within a class, traditional clustering methods often struggle to find a balance between the consensus level and the total adjustment cost, leading to challenges in effectively reaching a consensus and affecting the efficiency and accuracy of decision-making results.

(4) The insufficiency of feedback mechanisms: Many studies lack effective feedback adjustment mechanisms, relying on one-time or single-stage adjustments, failing to integrate diverse opinions at the individual and subgroup levels, leading to a lack of the consistency and stability in decision-making results.

1.4. Motivations and innovations

The above limitations highlight the urgent need for more advanced LSGDM methods. On account of this reason, we list main motivations below:

(1) The processing of incomplete and uncertain information requires superior techniques to accurately estimate missing data, thus enhancing the robustness of decision-making.

(2) The treatment of static trust relationships should be replaced by adaptive modeling to dynamically reflect changes in interactions among decision makers.

(3) The existing clustering methods are difficult to achieve an effective balance between consensus and adjustment costs, and more flexible algorithms are urgently needed to cope with diverse preferences.

(4) The lack of robust feedback mechanisms forces us to explore new consensus strategies to ensure the coherence and stability of decision-making at the individual and subgroup levels.

In view of this, the main innovations of this study can be summarized as follows:

(1) Neighborhood information estimation for missing values: In response to missing values in IFPRs, we exhibit a method that uses neighborhood information for estimation. This method effectively fills in the missing preference information by analyzing the similarity and trust relationships between DMs, thereby improving the completeness and accuracy of decision-making. Compared with some classic methods in the context of trust networks, these classic methods usually do not consider the K-nearest neighbors. Therefore, our method makes full use of the trust information of DMs, enhancing the integrity and accuracy of the decision-making process.

(2) Dynamic trust network construction and updating: We design a dynamic updating mechanism for the trust network to determine the trust degree of DMs and subgroups in real-time. This mechanism takes into account the historical performances of DMs, the similarity of current opinions, and other relevant factors, dynamically adjusting the structure of the trust network to ensure that the model can quickly adapt to environmental changes.

(3) Optimization of k-means clustering algorithm: In the CRP, we use an optimized k-means clustering algorithm that considers both preferences and preference adjustment costs. Our optimized algorithm balances the precision and cost of preference adjustment to optimize GDM results, effectively enhancing the robustness and efficiency of the decision-making process. Traditional clustering algorithms are mainly based on single-factor k-means integrated distance measurement, such as only considering the difference in preference relationships. Our algorithm considers more comprehensive factors and is closer to real situations, providing a new method for achieving an efficient consensus in real-world decision-making.

(4) Two-stage three-way feedback adjustment mechanism: To further improve the quality and stability of decision-making results, we introduce a two-stage feedback adjustment mechanism for individuals and subgroups, integrating the concept of TWD to facilitate consensus achievement. The first stage adjusts the opinions of individual DMs, and the second stage comprehensively adjusts the opinions at the subgroup level, ensuring the coordination and consistency of the overall decision-making plan. Compared with general consensus-reaching mechanisms, our method not only reaches a consensus at the global level but also reaches a consensus within each subgroup, enhancing the efficiency of the consensus. However, general consensus mechanisms only consider a global consensus.

For enhanced clarity, Table 1 presents a glossary of symbols along with their respective definitions.

Table 1
The description of mathematical symbols.

Mathematical symbols	Meaning
$E = \{e_1, e_2, \dots, e_p\}$	The set of DMs
$X = \{x_1, x_2, \dots, x_n\}$	The set of alternatives
$T = (t_{u,v})_{p\times p}$	The STN of DMs
$F_u = (f_{ij}^u)_{n\times n}$	The FPR matrix of the DM e_u
$F_{(l)} = (f_{ij}^{(l)})_{n\times n}$	The FPR matrix of the subgroup C_l
$\overline{F} = (\overline{f}_{ij})_{n\times n}$	The FPR matrix of the group
$UAC = \{UAC_1, UAC_2, \dots, UAC_p\}$	The set of unit adjustment cost of the DMs
e_u^a	The ath neighbor of the DM e_u
$e_u^{a[b]}$	The bth neighbor of the DM e_u^a
$F_{u,a} = (f_{ij}^{u,a})_{n\times n}$	The FPR matrix of the e_u^a
μ_u^a	The fuzzy membership degree of the DM e_u^a to another DM e_u
sim	The similarity of fuzzy preference information
r_{ij}^u	The reliability of the estimated missing value f_{ij}^u
$In_t(e_u)$	The trust influence of the DM e_u
θ_t	The threshold of trust influence
D	The trust discrepancy of DMs
θ_{ideal}	The threshold of consensus
d	The comprehensive difference of DMs
$C = \{C_1, C_2, \dots, C_k\}$	The set of subgroups
β	The control parameter for unit adjustment cost
Pos	The positive region
Bnd	The boundary region
Neg	The negative region
λ	The partition parameters for boundary and negative regions

1.5. Structure of the paper

The structure of this paper is arranged as follows: Section 2 begins by elucidating key concepts such as LSGDM, additive consistency, IFPRs, and STNs, thereby establishing a solid theoretical foundation for the research presented in this paper. Subsequently, Section 3 provides a detailed description of the proposed methods, including the KNN based estimation for missing values and the k-means clustering approach grounded in social trust networks. Building upon these methodologies, Section 4 further explores the consensus-reaching mechanism with minimum adjustment cost based on TWD. Section 5 demonstrates the practical application and effectiveness of the proposed methods through a specific case study, and conducts a sensitivity analysis of key parameters to assess the stability and reliability of the models. Section 6 further validates the applicability and superiority of the proposed methods through comparative analysis. Based on an in-depth analysis of the research findings, this paper discusses the theoretical and practical significance of the study. Ultimately, Section 7 summarizes the main contributions of this research and, based on the current study's outcomes, provides an outlook on potential future research directions, offering new ideas and avenues for subsequent studies.

To provide a clearer depiction of the proposed model, Fig. 1 illustrates the framework for an incomplete SN-LSGDM process.

2. Preliminaries

This section reviews some essential concepts, such as LSGDM, fuzzy sets, FPRs, IFPRs, additive consistency, STNs and so on.

2.1. LSGDM and FPRs

An LSGDM problem can be described as: a panel of DMs $E = \{e_1, e_2, \dots, e_p\}$ makes decisions on a set of alternatives $X = \{x_1, x_2, \dots, x_n\}$ according to certain criteria to select the best alternative x^*. FPRs are a commonly used evaluation form in LSGDM problems. However, missing evaluation information, known as IFPRs, often occurs in an actual decision-making. FPRs are essentially an application of fuzzy sets, expressing the degree of membership of elements within a set through the degree of membership provided by fuzzy sets.

In the real world, many issues are fraught with uncertainty and ambiguity, and traditional precise sets often struggle to accurately describe these problems. It is against this backdrop that the concept of fuzzy sets emerged, proposed by Zadeh (1965).

Definition 2.1 (*Zadeh, 1965*)**.** A fuzzy set A in a universe of discourse X is characterized by a membership function $\mu_A : X \to [0, 1]$, where for every element x in X, the value $\mu_A(x)$ represents the grade of membership of x in A. The membership function maps each element to a real number in the interval [0, 1], with 0 indicating no membership and 1 indicating full membership.

Preference relations are one of the common methods for evaluating alternatives. Two common types of preference relations include FPRs and IFPRs. These relations help DMs express their relative preferences among different alternatives. An FPR is used to deal with the uncertainty and fuzziness of DMs in the GDM, which is typically constructed using the concepts of fuzzy mathematics.

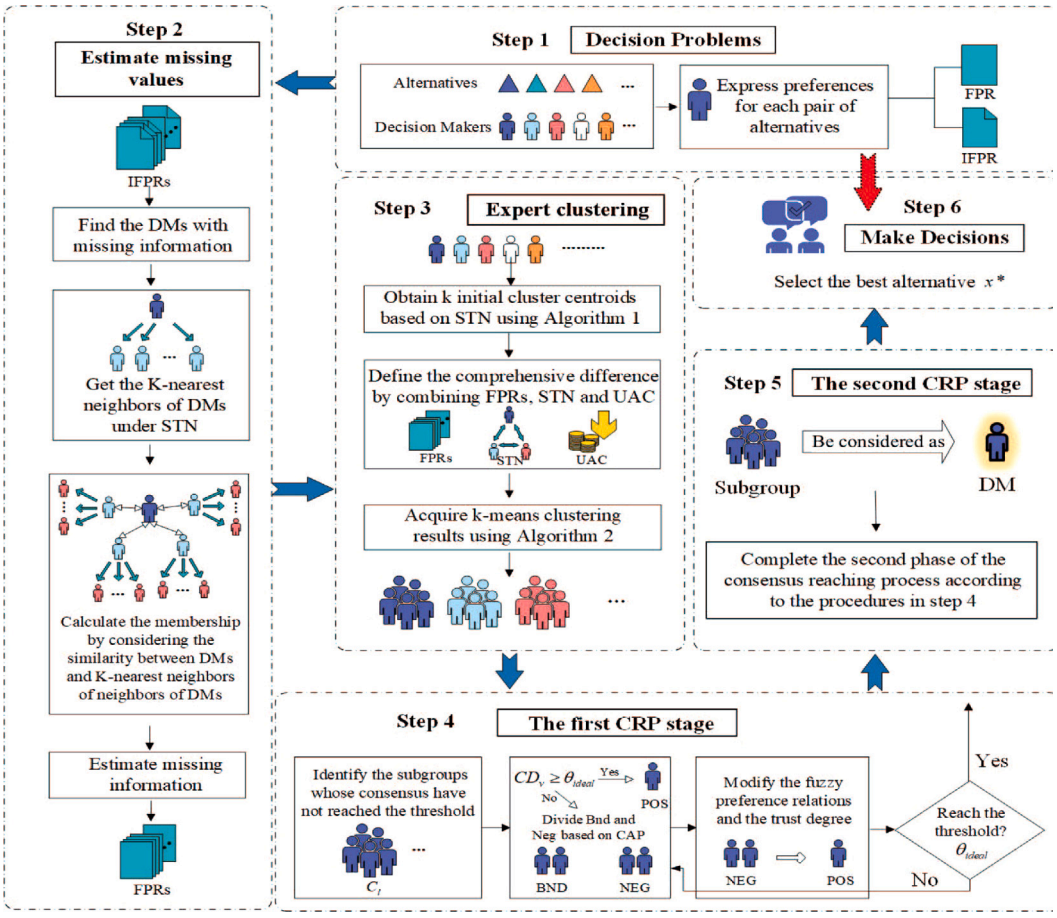


Fig. 1. Framework for SN-LSGDM under IFPR conditions.

Definition 2.2 (*Orlovsky, 1978*). An FPR matrix is defined as $F = (f_{ij})_{n\times n}$, where $f_{ij} \in [0,1]$ represents the degree of preference of DMs for the alternative x_i over the alternative x_j, and $f_{ij} + f_{ji} = 1, f_{ii} = 0.5$.

1. When $f_{ij} = 0.5$, it indicates that the alternatives x_i and x_j have no clear advantage or disadvantage for the DM, meaning they are equivalent ($x_i \sim x_j$).
2. When $0 \le f_{ij} < 0.5$, it indicates that the alternative x_j is relatively more preferred than the alternative x_i ($x_i < x_j$).
3. When $0.5 < f_{ij} \le 1$, it indicates that the alternative x_i is more favored than the alternative x_j, i.e., x_i is more preferred by the DM than x_j ($x_i > x_j$).

An IFPR refers to a situation where some elements of the FPR matrix are unknown. In practical applications, due to various reasons (such as incomplete information or DMs failing to provide clear preferences), some elements of the FPR matrix may be indeterminate. For these unknown elements, the symbol χ is used to denote the missing value.

Definition 2.3 (*Xu, Cabrerizo, & Herrera-Viedma, 2017*). Let $F = (f_{ij})_{n\times n}$ be an FPR. An FPR satisfies additive consistency if and only if

$$f_{ik} + f_{kj} = f_{ij} - 0.5, \quad \forall i, j, k = 1, 2, \dots, n. \tag{1}$$

The additive consistency has the advantages of being simple to implement and widely applied in FPRs, effectively enhancing the reliability of the decision-making process and the accuracy of the results. It provides a solid theoretical foundation for ensuring the consistency of preference relations, significantly improving decision quality. However, in most cases, FPRs do not naturally satisfy additive consistency.

Definition 2.4 (*Xu et al., 2017*). Let $F = (f_{ij})_{n\times n}$ be an FPR. An FPR $\tilde{F} = (\tilde{f}_{ij})_{n\times n}$ that fully satisfies an additive consistency can be constructed by the following formula:

$$\tilde{f}_{ij} = \frac{\sum_{k=1}^{n} (f_{ik} + f_{kj})}{n} - 0.5. \tag{2}$$

Definition 2.5 (*Xu et al., 2017*). Let $F = (f_{ij})_{n\times n}$ be an FPR and $\tilde{F} = (\tilde{f}_{ij})_{n\times n}$ be an FPR corresponding to F that satisfies the additive consistency. Then, the additive consistency degree of F is defined as:

$$ACD(F) = 1 - \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} |f_{ij} - \tilde{f}_{ij}|. \tag{3}$$

In general, an FPR with an additive consistency degree of at least 0.9 is considered to have an acceptable level of the additive consistency. Although the additive consistency is important within FPRs, it is not the main focus of this article. Therefore, in this article, we assume that the additive consistency of the given FPRs is acceptable.

Definition 2.6 (*Li et al., 2023*). Let the weight of each DM within the subgroup be equal with $\varphi_{(o)}^{u} = \frac{1}{q_o}$. The preference information of the o-th subgroup ($1 \le o \le k$) can be defined as $F_{(o)} = (f_{ij}^{(o)})_{n\times n}$, where $f_{ij}^{(o)}$ is described as

$$f_{ij}^{(o)} = \sum_{e_u \in C_o} \varphi_{(o)}^{u} f_{ij}^{u}, \quad e_u \in C_o. \tag{4}$$

Let the weight of each subgroup be equal with $\phi_{(o)} = \frac{q_o}{p}$. The group preference information is defined as $\overline{F} = (\overline{f}_{ij})_{n\times n}$, where $\overline{f}_{ij}$ can be obtained as:

$$\overline{f}_{ij} = \sum_{C_o \in C} \phi_{(o)} f_{ij}^{(o)}. \tag{5}$$

Definition 2.7 (*Herrera-Viedma, Martinez, Mata, & Chiclana, 2005; Liu, Liang, Chiclana, & Wu, 2017*)**.** The consensus degree can be defined at three levels: DMs, subgroups as well as the group.

Level 1: The consensus degree between the DM e_u and the subgroup C_o is defined as:

$$CD_u = 1 - \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} \left| f_{ij}^{u} - f_{ij}^{(o)} \right|. \tag{6}$$

Level 2: The intra-group consensus degree of the subgroup C_o can be calculated as:

$$CD_{(o)} = 1 - \sum_{e_u \in C_o} \varphi_{(o)}^{u} \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} \left| f_{ij}^{u} - f_{ij}^{(o)} \right|. \tag{7}$$

Level 3: The group consensus degree can be obtained as follows:

$$GCD = 1 - \sum_{C_o \in C} \phi_{(o)} \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} \left| f_{ij}^{(o)} - \overline{f}_{ij} \right|. \tag{8}$$

Definition 2.8 (*Ben-Arieh & Easton, 2007*)**.** Let $UAC = \{UAC_1, UAC_2, \ldots, UAC_p\}$ be the UAC for a group of DMs $E = \{e_1, e_2, \ldots, e_p\}$, where $0 < UAC_u \leq 1, 1 \leq u \leq p$.

The UAC, which ranges between 0 and 1, serves as an indicator of relative magnitude, reflecting the relative difficulty of adjustments rather than a specific numerical value.

2.2. Social trust networks (STNs)

STN is a powerful tool for measuring the trust relationships among DMs, where this concept has been widely applied across various fields since its introduction by Wasserman and Faust (1994) in their 1994 publication. STN typically represents the structure of networks in three main forms: sociometry, algebraic methods, and graph theory. Although traditional sociometry is intuitive in identifying relationships within networks, it has limitations when dealing with the nuances of trust intensity.

To overcome this limitation, researchers have developed fuzzy sociometry, which allows the measurement of trust relationships to be a continuous value between 0 and 1, rather than a simple presence or absence, thus more accurately reflecting the trust intensity between DMs. For instance, the concept of fuzzy sociometry was proposed by Dong et al. (2018) and Pérez, Mata, and Chiclana (2014), which takes into account the varying degrees of trust, providing a richer perspective for analyzing and interpreting trust relationships in social networks. In this way, STN can more delicately capture and express the complex and dynamic interactions between DMs.

Definition 2.9 (*Dong et al., 2018*)**.** Let $T = (t_{u,v})_{p\times p}$ be the trust network relation matrix of the group of DMs E, where $t_{u,v}$ is the trust degree of the DM e_u in the DM e_v, satisfying $0 \leq t_{u,v} \leq 1, t_{u,u} = 1$.

2.3. K-Nearest neighbors (KNNs)

The KNN algorithm is a simple and intuitive non-parametric method used for both classification and regression. In classification tasks, the algorithm calculates the distance between a sample and each sample in the training dataset to identify the K-nearest neighbors. It then determines the category of the sample based on a majority vote of the categories of these K-nearest neighbors. In regression tasks, the KNN algorithm also finds K-nearest neighbors but predicts the output value of the sample by averaging the values of these neighbors. The K-nearest neighbors can effectively reflect local information and make full use of the patterns and relationships in the existing data, making the estimation of missing values more accurate and reliable.

Definition 2.10 (*Cover & Hart, 1967*)**.** Given a group of DMs $E = \{e_1, e_2, \ldots, e_p\}$, let $dis(\cdot,\cdot)$ be the distance between two DMs under a given metric. Then, the K-nearest neighbor of the DM e_u under the given metric is defined as

$$N_K(e_u) = \{e_v \mid dis(e_u, e_v) \leq dis_K, v \neq u\}, \tag{9}$$

where dis_K is the distance between the DM e_u and the K-nearest neighbor.

Remark 2.1. The K for K-nearest neighbors in this paper is represented in an uppercase. In contrast, the k in the k-means clustering algorithm is represented in a lowercase.

Definition 2.11. The final score $Score_i$ for each alternative x_i is

$$Score_i = \sum_{j=1}^{n} \overline{f}_{ij}, \tag{10}$$

where $\overline{F} = (\overline{f}_{ij})_{n\times n}$ is the final group opinion.

3. Missing values estimation in IFPRs and SN clustering

The section focuses on techniques for missing value estimation and SN clustering. In Section 3.1, a method for missing value processing that integrates the KNN algorithm with the theory of fuzzy memberships is introduced, aiming to improve the completeness of the dataset. Section 3.2 presents a k-means clustering method based on STNs, which uses the structural features of SNs to enhance the clustering effect.

3.1. Missing value estimation based on KNNs and fuzzy membership degrees

In dealing with IFPRs, we employ a refined version of the traditional KNN method to meet the specific demands of STNs. Within the STN, DMs' opinions are often influenced by the opinions of other DMs they trust. Thus, we focus on identifying the K-nearest neighbors in the STN who have a high level of trust with the target DM. The preference relations of these neighbors serve as a crucial reference for estimating the missing values, thereby aiding in the more accurate inference of the absent preference information.

Definition 3.1. Let T be the trust network relations of the group of DMs E and F be an IFPR. For a DM e_u with an missing value f_{ij}^{u} in the element of the IFPR matrix, the K-nearest neighbor set in an STN is defined as:

$$N_T^K(e_u) = \{e_v \mid t(e_u, e_v) \geq t(e_u, e_u^K), f_{ij}^{v} \neq \chi\}, \tag{11}$$

where e_u^K is the Kth nearest neighbor of e_u in an STN, and e_v is considered as a neighbor, based on the matrix where the element f_{ij} is known.

Remark 3.1. In the event of an extreme situation, such as being unable to find the required K-nearest neighbors that meet the conditions, this study proposes to use the average of all DMs' ratings on that attribute as the estimated value, thereby ensuring that the information loss in the decision-making system is minimized.

After identifying the K-nearest neighbors, we introduce the concept of fuzzy membership degrees to represent the reference value of each neighbor when DMs estimate missing values, which can also be understood as the similarity between DMs and their neighbors. To further determine the reference value of each neighbor, we take into account the similarity and trust degree between DMs and K-nearest neighbors of DMs, and then weight them accordingly. The definition of this fuzzy membership degree not only reflects the similarity of fuzzy preferences but also the depth of trust relationships. To distinguish clearly, let e_u^a represent the ath neighbor of the DM e_u, and $e_u^{a[b]}$ represent the bth nearest neighbor of the ath neighbor e_u^a of the DM e_u, where $1 \le a \le K, 0 \le b \le K$. Specifically, $e_u^{a[0]}$ denotes e_u^a.

Definition 3.2. The similarity of fuzzy preference information for e_u and $e_u^{a[b]}$ regarding the alternatives x_i, x_j is calculated as follows:

$$s(e_u, e_u^{a[b]}) = 1 - \frac{\sum_{j \in KI_{i,\cdot}} |f_{ij}^u - f_{ij}^{u,a[b]}| + \sum_{i \in KI_{\cdot,j}} |f_{ij}^u - f_{ij}^{u,a[b]}|}{|KI_{i,\cdot}| + |KI_{\cdot,j}|}, \tag{12}$$

where $KI_{i,\cdot}$ represents the set of alternatives for which both DM e_u and DM $e_u^{a[b]}$ have known fuzzy preference information regarding the alternative x_i, that is, $KI_{i,\cdot} = \{x_j \mid f_{ij}^u$ and $f_{ij}^{u,a[b]}$ have known$\}$, and $KI_{\cdot,j}$ represents the set of alternatives for which both DM e_u and DM $e_u^{a[b]}$ have known fuzzy preference information regarding the alternative x_j, that is, $KI_{\cdot,j} = \{x_i \mid f_{ij}^u$ and $f_{ij}^{u,a[b]}$ have known$\}$.

Definition 3.3. Consider the similarity of the fuzzy preference information of in the K-nearest neighbors $e_u^{a[b]} \in N_T^K(e_u^a)$ under the STN. The calculation formula is as follows:

$$sim_K(e_u, e_u^a) = \frac{\sum_{b=0}^{K} s(e_u, e_u^{a[b]})}{K+1}, \tag{13}$$

where $e_u^{a[0]} = e_u^a$.

Under normal circumstances, if there are missing elements f_{ij}^u in the preference relation matrix of the DM e_u, the missing values could be estimated by referring to the values of DMs who have high similarity and high credibility with the DM e_u at those positions.

Definition 3.4. In accordance with the KNN principle, the process of identifying the K DMs with the most reliable f_{ij} values and the highest trustworthiness, denoted as $e_u^1, \ldots, e_u^K$, is defined. Subsequently, the fuzzy membership degree of a DM e_u with respect to each of these K DMs, e_u^a (where $1 \le a \le K$), is calculated based on their available fuzzy preference information. The computation of this fuzzy membership degree is governed by the following formula:

$$\mu_u^a = \frac{t + sim_K(e_u, e_u^a)}{1 + sim_K(e_u, e_u^a)}, \tag{14}$$

where $sim_K(e_u, e_u^a)$ represents the similarity of the known fuzzy preference information between two DMs, and t is the trust of the DM e_u towards the DM e_u^a.

By performing normalization on μ_u^a, we can obtain:

$$\widetilde{\mu}_u^a = \frac{\mu_u^a}{\sum_{a=1}^{K} \mu_u^a}. \tag{15}$$

Using these fuzzy membership degrees as weights, we proceed to impute missing values. The specific steps include: first, calculating the fuzzy membership degrees for the K-nearest neighbors corresponding to each missing value, then using these weights to sum the known preference values of the neighbors to estimate the missing value. To improve the accuracy of the estimation, this process can be iterated, updating the missing values and membership degrees after each iteration until the results converge.

In practical applications, the order of estimation significantly affects the outcome. We propose a strategy to prioritize the estimation of missing values that are associated with a higher number of known values, as these are more likely to be accurately estimated and provide a reliable reference for the estimation of other missing values. Additionally, the order of estimation can be arranged based on the magnitude of fuzzy membership degrees, addressing the missing values with higher degrees first to gradually strengthen the accuracy and stability of the entire estimation process.

In an incomplete system, this method of estimating missing values based on the fuzzy membership degree of K-nearest neighbors in the STN will encounter a situation where updating a missing value will affect the membership relations of other DMs with missing values. Therefore, modify the most confident missing value first.

Definition 3.5. Select the situation with the greatest confidence first, and then the reliability for the missing information f_{ij}^u can be computed as:

$$r_{ij}^u = \sum_{a=1}^{K} \mu_u^a. \tag{16}$$

Therefore, in each modification, first modify the missing values with the highest reliability, then update the other missing values' fuzzy membership degrees, and proceed with the modifications until all missing values are estimated.

Definition 3.6. The formula for estimating in the missing values is as follows:

$$f_{ij}^u = \sum_{a=1}^{K} \widetilde{\mu}_u^a f_{ij}^{u,a}, \tag{17}$$

where $F_{u,a} = (f_{ij}^{u,a})_{n \times n}$ is the FPR of the DM e_u^a.

Additionally, the estimation formula for the symmetric position of f_{ij}^u, which is f_{ji}^u, is:

$$f_{ji}^u = \sum_{a=1}^{K} \widetilde{\mu}_u^a f_{ji}^{u,a}. \tag{18}$$

It is evident that due to the properties of the FPR, we can easily deduce:

$$f_{ij}^u + f_{ji}^u = 1. \tag{19}$$

Remark 3.2. The process of estimating missing values has a time complexity of $O(K^2np)$.

3.2. The k-means clustering method based on STNs

With the development of big data and machine learning technologies, clustering algorithms are widely applied across various fields and industries. For instance, Borlea, Precup, Dragan, and Borlea (2017) proposed a new centroid update method for the k-means clustering algorithm, which reduced the number of iterations by updating centroids at each step of the algorithm, thereby accelerating the clustering process, and its effectiveness has been validated on multiple datasets. Recently, many scholars have made significant efforts in the fields of clustering and optimization. The KM-SML method proposed by Borlea, Precup, and Borlea (2022) enhances k-means clustering by detecting and correcting misclassifications with supervised learning, particularly improving results on the Iris dataset. Furthermore, Kilic, Sarac Essiz, and Kaya Keles (2023) designed a new binary anarchistic social optimization algorithm for feature selection and verified its effectiveness and superiority on multiple datasets.

In this paper, we design an improved k-means clustering method based on STNs, which is an algorithm designed to categorize DMs who have mutual trust, similar preferences, and similar adjustment costs into the same group. This method addresses the limitations of parameter selection in traditional k-means algorithms, especially the

determination of the k value (the number of clusters), by proposing a strategy for adaptively determining k, using comprehensive influence as a measurement standard, replacing the original Euclidean distance. Our proposed algorithm adaptively determines the optimal k value by analyzing the influence and trust relationships of nodes (DMs) in the STN. This method does not rely on prior knowledge and adjusts dynamically to ensure that the number of clusters matches the characteristics of the data and actual needs, thereby improving the stability and practicality of the algorithm.

Definition 3.7. Consider the trust of the DM team E towards the DM e_u, referred to as the trust influence of the DM e_u, the calculation formula is as follows:

$$In_t(e_u) = \frac{\sum_{v=1}^{p} t_{v,u}}{p}. \tag{20}$$

The selection of initial cluster centroids should ideally have high trust influence and low mutual trust. Set the threshold as:

$$\theta_t = std_t(T) = \sqrt{\frac{\sum_{u=1}^{p}\sum_{v=1}^{p}(t_{u,v} - \bar{t})^2}{p^2}} \tag{21}$$

$$\bar{t} = \frac{\sum_{u=1}^{p}\sum_{v=1}^{p} t_{u,v}}{p^2}. \tag{22}$$

Let the discrepancy between e_u and e_v as

$$D = 1 - \frac{t_{u,v} + t_{v,u}}{2}. \tag{23}$$

The Algorithm 1 for adaptively selecting cluster centroids is shown below:

Algorithm 1: The algorithm for initializing k cluster centroids adaptively

Input: The DM set E and the social trust network T
Output: Initial cluster centroids set $S = \{s_1, s_2, \cdots, s_k\}$

Initialize set $S = \emptyset$;
Calculate the trust influence in_t by Eq. (20) of each DM e_u;
Calculate the threshold θ_t by Eq. (21);
Sort the DMs in a descending order of the trust influence In_t and denote the resulting order as $E_{order} = \{e_{order_1}, e_{order_2}, \cdots, e_{order_p}\}$;
$S = S + \{e_{order_1}\}$;
while E_{order} *is not empty* **do**
    Calculate the minimum discrepancy D by Eq. (23) between the trust influence of the first item in E_{order} and each DM in S;
    if $D \le \theta_t$ **then**
        Delete the first item in E_{order};
    end
    else
        Add the first item of the set E_{order} to the set S and remove it from the set E_{order};
    end
end
Output the initial cluster centroids set $S = \{s_1, s_2, \cdots, s_k\}$.

Remark 3.3. The time complexity of Algorithm 1 is $O(p)$.

In the traditional k-means algorithm, the similarity between data points is mainly measured by Euclidean distance. However, in LSGDM, this simple measurement method is insufficient to fully reflect the complex trust relationships and preference similarities. To address this issue, we have introduced a new measurement method, which combines the trust and preference similarity between DMs to improve the completeness and accuracy of decision-making.

This section delves into the quantification of mutual influence within a decision-making framework, where the interactions among DMs, between DMs and subgroups, and between subgroups are meticulously examined. The metrics for these influences are grounded in the community trust network and complete FPRs, providing a nuanced view of the dynamics at play.

Level 1: The mutual influence between DMs is a metric derived from the community trust network and complete FPRs that quantifies the reciprocal impact or interaction between pairs of DMs. The specific calculation of this metric is detailed in the following formula:

$$MI(e_u, e_v) = \frac{t_{u,v} + t_{v,u}}{1 + \|F_u - F_v\|_F^2}, \tag{24}$$

where $F_u = (f_{ij}^u)_{n\times n}$, $F_v = (f_{ij}^v)_{n\times n}$.

Level 2: The mutual influence between a DM and a subgroup is a measure that reflects the impact of the interactions between a single DM, denoted as e_u, and a collective of DMs within a subgroup C_l, based on the community trust network and complete FPRs. The specific calculation is formalized as follows:

$$MI(e_u, C_l) = \frac{t_{u,(l)} + t_{(l),u}}{1 + \|F_u - F_{(l)}\|_F^2}, \tag{25}$$

where $t_{u,(l)} = \frac{\sum_{e_v\in C_l} t_{u,v}}{q_l}, t_{(l),u} = \frac{\sum_{e_v\in C_l} t_{v,u}}{q_l}$, q_l is the number of DMs in the subgroup C_l and $F_{(l)} = (f_{ij}^{(l)})_{n\times n}, f_{ij}^{(l)} = \frac{\sum_{e_v\in C_l} f_{ij}^v}{q_l}$.

Level 3: The mutual influence between subgroups is a quantitative measure that captures the level of interaction and influence between two distinct subgroups of DMs, C_o and C_l, within the framework of a community trust network and informed by complete FPRs. The calculation of this mutual influence is defined by the following formula:

$$MI(C_o, C_l) = \frac{t_{(o),(l)} + t_{(l),(o)}}{1 + \|F_{(o)} - F_{(l)}\|_F^2}, \tag{26}$$

where $t_{(o),(l)} = \frac{\sum_{e_u\in C_o, e_v\in C_l} t_{u,v}}{q_o q_l}, t_{(l),(o)} = \frac{\sum_{e_v\in C_l, e_u\in C_o} t_{v,u}}{q_l q_o}$, q_o, q_l are the numbers of DMs in the subgroups C_o, C_l, respectively. $F_{(o)} = (f_{ij}^{(o)})_{n\times n}, f_{ij}^{(o)} = \frac{\sum_{e_u\in C_o} f_{ij}^u}{q_o}$.

Remark 3.4. For ease of calculation, normalize all mutual influences between DMs to a scale of 0–1, this paper uniformly uses $Y(\cdot) = \frac{\cdot - \min}{\max - \min}$.

In addition, we replace the Euclidean distance in the traditional k-means algorithm with comprehensive influence. This comprehensive influence not only considers the preference similarity of DMs but also includes their status in the network and their influence on other DMs. This measurement method allows the clustering results to more truly reflect the mutual relationships between DMs.

Next, we introduce the concept of comprehensive difference, a key metric for evaluating disparities in decision-making contexts. This metric integrates the UAC of DMs and the effects of mutual influence, providing a holistic view of the interactions within and between groups. The levels that follow present the formulas for calculating these differences.

Level 1: The comprehensive difference between two DMs, incorporating UAC and mutual influence, is detailed through the subsequent formula:

$$d(e_u, e_v) = \frac{\left(1 - MI(e_u, e_v)\right) + |UAC_u - UAC_v|}{2}. \tag{27}$$

This measure is designed to assess the distinction between the preferences and influences of two DMs within a decision-making framework. Similarly, the comprehensive difference between a DM and a subgroup, and between subgroups, can be defined.

Level 2: The formula for the comprehensive difference between a DM and a subgroup is as follows:

$$d(e_u, C_l) = \frac{\left(1 - MI(e_u, C_l)\right) + |UAC_u - UAC_{(l)}|}{2}, \tag{28}$$

where $UAC_{(l)} = \frac{\sum_{v \in C_l} UAC_v}{q_l}$.

Level 3: The calculation of the comprehensive difference between two subgroups is presented below:

$$d(C_o, C_l) = \frac{\left(1 - MI(C_o, C_l)\right) + |UAC_{(o)} - UAC_{(l)}|}{2}, \tag{29}$$

where $UAC_{(o)} = \frac{\sum_{u \in C_o} UAC_u}{q_o}$.

To further improve the clustering effect, we have proposed a method for adaptively selecting initial cluster centers. As shown in Algorithm 2, this method uses information from the STN to select nodes that are in important positions in the trust network and have a significant influence on other DMs as initial cluster centers, thereby guiding the clustering process and enhancing the quality and stability of the clustering results.

Algorithm 2: The improved k-means algorithm based on STN

Input: The DM set E, the social trust network T, the complete fuzzy preference relation FPR, the initial cluster centroids set S, the unit adjustment cost UAC and the maximum iteration τ_{max}

Output: Clustering result $C = \{C_1, C_2, \cdots, C_k\}$

Initialize iteration $t = 0$, cluster $C^0 = \{C_1^0, C_2^0, \cdots, C_k^0\}$, where $C_i^0 = [s_i^0]$, $SSE^0 = 0$, the difference of SSE between two iterations $\Delta_{SSE} = 1$;
repeat
  Calculate the comprehensive difference $d(e_u, e_v)$ by Eq. (27) from each DM to each cluster centroid $d(e_u, s_i^t)$;
  $t = t + 1$;
  Assign each DM to the cluster with the smallest comprehensive difference to obtain a new clustering result C^t;
  Calculate the comprehensive difference $d(e_u, C_o)$ by Eq. (28) from each DM to its cluster;
  Calculate SSE^t by Eq. (30);
  Calculate $\Delta_{SSE} = \left|SSE^t - SSE^{t-1}\right|$;
  Update the cluster centroid of each cluster, select the one with the smallest comprehensive difference to the cluster as the new cluster centroid s_i^t;
  Calculate SSE^t
until $\Delta_{SSE} \to 0$ *or* $t \geq \tau_{max}$;
Output the clustering result $C = C^t = \{C_1, C_2, \cdots, C_k\}$.

Remark 3.5. The time complexity of Algorithm 2 is $O(knp^2)$.

To ensure the efficiency and stability of the algorithm, the termination condition for the clustering process is the convergence of the sum of square Error (SSE).

Definition 3.8. The sum of square error, abbreviated as SSE, is a measure that aggregates the squared distances of data points within a cluster to their respective class center. The calculation formula is as follows:

$$SSE = \sum_{u=1}^{p} \left(d(e_u, C_o)\right)^2, \tag{30}$$

where C_o is the subgroup to which the DM e_u belongs.

In summary, the improved k-means clustering method based on STNs achieves more efficient and accurate clustering of DMs by adaptively determining k and selecting initial cluster centers, using comprehensive influence as a measurement indicator. This method not only overcomes the shortcomings of the traditional k-means algorithm but also better deals with the complexity and uncertainty in LSGDM.

4. A consensus reaching mechanism with minimal adjustment cost based on TWD

This section introduces a consensus reaching mechanism based on the TWD theory, which aims to minimize adjustment costs among DMs and employs a two-stage adaptive dynamic optimization strategy to address decision-making issues at different stages.

4.1. TWD criteria

In the TWD framework, we first identify and distinguish three key decision-making regions:

1. Pos: This set includes DMs whose opinions meet the consensus requirements and do not require any adjustments;
2. Neg: This set includes DMs who need to adjust their opinions;
3. Bnd: This is a region for deferred decision-making, where DMs neither meet the consensus requirements nor need to be modified immediately, but are reserved for subsequent observation.

To adapt to LSGDM in STNs, we have designed the decision-making regions for the three-way consensus process as follows: Taking the subgroup C_l as an example, Pos is composed of DMs whose opinion similarity exceeds the consensus threshold, and their opinions do not require modification. θ_{ideal} is the given consensus threshold we aim to achieve. The formula for calculating Pos is:

$$Pos = \{e_v \mid CD_v \geq \theta_{ideal}, e_v \in C_l\}. \tag{31}$$

In extreme cases, if there are no DMs in the Pos, the subgroup opinion is added as a virtual DM to the Pos, with the average UAC as the virtual DM's UAC and the average trust degree as the virtual DM's trust degree.

After determining the Pos, we need to divide the remaining DMs into the Bnd and the Neg. We hope that DMs in the Neg can be adjusted first. In the STN, we tend to use the opinions of other DMs with higher trust degrees as a reference for modification and hope that the adjustment cost is as low as possible. Therefore, we select DMs who have a larger difference with the group opinion, a lower UAC, and a higher trust degree towards the DMs in the Pos for opinion adjustment. For this, we define the concept of comprehensive adjustment priority (CAP).

The CAP is a measure used to evaluate the resistance to change of a DM's opinion within a subgroup, taking into account the DM's trust in the consensus and the cost associated with adjusting their opinion. CAP can guide the decision-making process and ensure that it is both economical and effective, while respecting and exploiting the trust relationship between DMs. By prioritizing integrated adjustments to high-priority DMs, the consensus can be reached more quickly, while maintaining the quality and efficiency of decision-making.

Definition 4.1. For a DM e_r belonging to a subgroup set C_l but not in the Pos (i.e., $e_r \in C_l - Pos$), the CAP_r is calculated as follows:

$$CAP_r = \beta \cdot (1 - UAC_r) + \frac{(1-\beta)}{2} \cdot t_r + \frac{(1-\beta)}{2} \cdot (1 - CD_r), \tag{32}$$

where $t_r = \frac{\sum_{e_v \in Pos} t_{r,v}}{|Pos|}$ and β is the parameter controlling the adjustment cost.

If the DM satisfies the case $CAP \geq CAP_\lambda$, the DM is categorized into the Neg; otherwise, the DM is categorized into the Bnd.

$$CAP_\lambda = CAP_{\min} + \lambda \left(CAP_{\max} - CAP_{\min}\right), \lambda \in [0, 1]. \tag{33}$$

The parameter λ is a value that ranges between 0 and 1, used to adjust the value of CAP_λ, so that it varies between $CAP_{\min}$ and $CAP_{\max}$. When $\lambda = 0$, CAP_λ is equal to $CAP_{\min}$, indicating the least urgent adjustment scenario. When $\lambda = 1$, CAP_λ is equal to $CAP_{\max}$, indicating

the most urgent adjustment scenario. This formula essentially performs a linear interpolation between $CAP_{\min}$ and $CAP_{\max}$ based on the value of λ. By adjusting λ, the Bnd between the Neg and the Bnd can be controlled, thereby influencing which DMs need to be prioritized for adjustment.

Foe instance, in emergency management, we need to quickly gather the opinions of experts. In this situation, we can set the parameter λ to 1 to control the negative and boundary domains, which can reduce the number of experts that need to be adjusted, thereby accelerating the response speed. For decision-making on infrastructure engineering, this usually requires calm and thoughtful consideration. In this case, we can set λ to 0, which will increase the number of DMs that need to be adjusted. Although this will increase costs, the consensus achieved in the end will also be higher. When it comes to strategic decisions involving the government or industry, we need to consider a variety of factors. We can choose to minimize the amount of adjustment or maximize the final consensus based on the actual situation. It is worth mentioning that our method can flexibly handle both scenarios.

Remark 4.1. The time complexity of a single optimization model is $O(n^2 p)$, which indicates the computational requirements when dealing with larger-scale data.

4.2. Two-stage adaptive dynamic consensus model in STNs

In the first stage, a subgroup is considered individually. In the second stage, each subgroup is regarded as a single DM, and all subgroups are treated as one large subgroup to reach a consensus using the method from the first stage.

It is clear that the larger the Pos, the higher the consensus. Therefore, considering the minimum adjustment cost, only the DMs in the Neg are changed. For DMs in the Bnd, the choice is to delay modification, which means that after dealing with the instances in the Neg, a decision is made on whether to modify the instances in the Bnd based on the situation. This strategy of deferred modification allows DMs to make more accurate decisions after obtaining more information or having a deeper understanding of the issue. In the first stage, taking the subgroup C_l as an example, an adaptive dynamic optimization model under the STN is presented, with the aim of minimizing adjustment costs as follows:

$$\min \sum_{e_u \in Neg} UAC_u \sum_{i,j=1}^{n} |h_{ij}^u - f_{ij}^u|$$

subject to

$$\begin{cases} Pos = \{e_v \mid CD_v \geq \theta_{ideal}, e_v \in C_l\}, \\ CAP_r = \beta \cdot (1 - UAC_r) + \dfrac{(1-\beta)}{2} \cdot t_r \\ \quad + \dfrac{(1-\beta)}{2} \cdot (1 - CD_r), e_r \in C_l - Pos, \\ CAP_\lambda = CAP_{\min} + \lambda \left(CAP_{\max} - CAP_{\min}\right), \\ Neg = \{e_u \mid CAP_u \geq CAP_\lambda\}, \\ Bnd = \{e_w \mid CAP_w < CAP_\lambda\}, \\ |h_{ij}^u - \sum_{e_v \in Pos} \widetilde{t}_v f_{ij}^v| \leq \epsilon, e_u \in Neg, \\ \widetilde{t}_v = \dfrac{t_{u,v}}{\sum_{e_v \in Pos} t_{u,v}}, e_u \in Neg. \end{cases} \tag{34}$$

where h_{ij}^u is a decision variable for all $e_u \in Neg, 1 \leq i,j \leq n, t_r = \frac{\sum_{e_v \in Pos} t_{r,v}}{|Pos|}$. Let f_{ij}^{u*} represent the value of the decision variable h_{ij}^u when the model (34) achieves its optimal solution. θ_{ideal} is the given consensus threshold we aim to achieve. $\sum_{e_v \in Pos} \widetilde{t}_v f_{ij}^v$ represents the adjustment value given by the trust in the DMs of the Pos for the DMs in the Neg who need to be modified, and ϵ is the allowable error. After obtaining the optimized adjustment cost, current CD is compared with the set threshold. If satisfied, the CRP ends, otherwise, update the trust values, recalculate the Pos, Neg, and Bnd, and repeat the above operations.

Trust Value Update Mechanism: The DMs in the Neg who utilize information from the Pos will correspondingly increase their trust in the DMs of the Pos. For the DM $e_u \in Neg$, its trust level in the DM $e_v \in Pos$ is updated as follows:

$$t_{u,v}^* = t_{u,v}^{1-t_{u,v}}. \tag{35}$$

The advantage of this formula is that the trust of the DM e_u in the DM e_v will increase. The preference information of the DMs in the Neg has changed. Naturally, the DMs in the Bnd will change their trust in the DMs in the Neg. Hence, the trust of DMs in the Bnd $e_w \in Bnd$ towards DMs in the Neg must also be updated:

$$t_{w,u}^* = t_{w,v}^{1-t_{w,v}}. \tag{36}$$

After the consensus is reached within each group, the second stage begins by first calculating the trust between subgroups (already given above) and the average UAC for each subgroup.

For this issue, an optimization algorithm implemented in Python is adopted. When dealing with constraints, a Linear Programming problem is formulated and the 'linprog' function is utilized to find a solution. Specifically, an objective function is defined, which aims to minimize the adjustment cost. Subsequently, a series of linear constraints are constructed to ensure that the solution found is within an acceptable range. By employing a linear programming solver, we can identify the optimal solution which satisfies all the constraints and minimizes the objective function.

During the iterative process, the algorithm continuously updates the preference relations of the DMs until the consensus level of the entire group reaches the θ_{ideal}. In this process, the algorithm also dynamically adjusts the trust network to facilitate the formation of consensus among different DMs. Overall, the algorithm optimizes the decision-making process by leveraging linear programming techniques while ensuring compliance with a set of predefined constraints.

Definition 4.2. The UAC of the subgroup C_l $(1 \leq l \leq k)$ can be calculated as follows:

$$UAC_{(l)} = \sum_{e_v \in C_l} \varphi_l^v UAC_v. \tag{37}$$

Adjust the group consensus degree GCD and compare it with the given θ_{ideal}. If greater, the CRP ends, otherwise, enter the dynamic feedback adjustment mechanism:

$$\min \sum_{C_o \in Neg} UAC_{(o)} \sum_{i,j=1}^{n} \left|h_{ij}^{(o)} - f_{ij}^{(o)}\right|$$

subject to

$$\begin{cases} Pos = \{C_l \mid CD_{(l)} \geq \theta_{ideal}, C_l \in C\}, \\ CAP_{(s)} = \beta \cdot (1 - UAC_{(s)}) + \dfrac{(1-\beta)}{2} \cdot t_{(s)} + \dfrac{(1-\beta)}{2} \cdot (1 - CD_{(s)}), \\ CAP_\lambda = CAP_{\min} + \lambda \left(CAP_{\max} - CAP_{\min}\right), \\ Neg = \{C_o \mid CAP_{(o)} \geq CAP_\lambda\}, \\ Bnd = \{C_y \mid CAP_{(y)} < CAP_\lambda\}, \\ \left|h_{ij}^{(o)} - \sum_{C_l \in Pos} \widetilde{t}_{(l)} f_{ij}^{(l)}\right| \leq \epsilon, C_o \in Neg, \\ \widetilde{t}_{(l)} = \dfrac{t_{(o),(l)}}{\sum_{C_l \in Pos} t_{(o),(l)}}, C_o \in Neg, \end{cases} \tag{38}$$

where $h_{ij}^{(o)}$ is a decision variable for all $C_o \in C, 1 \leq i,j \leq n, t_{(s)} = \frac{\sum_{C_l \in Pos} t_{(s),(l)}}{|Pos|}$. Let $f_{ij}^{(o)*}$ represent the value of the decision variable $h_{ij}^{(o)}$

when the model (38) achieves its optimal solution. Similar to the first phase, after optimization, compare the consensus degree with the threshold. If greater, the CRP ends, otherwise, update the trust values, then recalculate the Pos and Bnd, and optimize again until consensus is reached.

The formula for updating trust values: Similar to the first phase, the subgroup in the $Neg\ C_o$ gets closer to the subgroup in the $Pos\ C_l$, so the trust level of subgroup C_o in the subgroup C_l will be higher:

$$t^*_{(o),(l)} = t^{1-t_{(o),(l)}}_{(o),(l)}. \tag{39}$$

Similarly, the trust level of the subgroup in the $Bnd\ C_y$ towards the subgroup in the $Neg\ C_o$ is updated as:

$$t^*_{(y),(o)} = t^{1-t_{(y),(o)}}_{(y),(o)}. \tag{40}$$

The advantages of this optimization method are as follows:

(1) Unlike traditional optimization functions that minimize adjustment costs, this method provides a reference range for the DMs who need to be adjusted (those in the Neg) based on the trust network, which is in line with the actual decision-making process and makes the entire CRP more convincing.

(2) The consensus process is also a process of enhancing trust. After one adjustment, the DMs in the Neg trust those in the Pos more, and DMs in Bnd also enhance their trust in those in the Neg.

(3) By fully leveraging the strengths of TWD, a deferred modification is made for the DMs in the Bnd. After the preference information of the DMs in the Neg has been adjusted, a determination is made based on the actual situation. This makes the entire CAP more conclusive and avoids the waste of resources to a certain extent.

Remark 4.2. The time complexity of the entire CRP is $O(kn^2p)$. Considering all parts of the algorithm, the overall time complexity of the model can be expressed as $O(K^2pn + p + knp^2 + kn^2p)$. In practical problems, taking into account that K is relatively small and $p > n$, the final time complexity can be simplified to $O(knp^2)$.

5. An illustrative example

This section demonstrates the implementation steps of the solution through a case study of diversified investment and verifies the effectiveness of the adopted method.

5.1. Problem description

In this section, we show a numerical example to verify the execution steps and validity of the proposed model. This case study refers to the design in Gong et al. (2024) and aims to illustrate a specific application of venture capital. For more details, see https://github.com/JustinaZhan/Datasets-and-code. To achieve optimal diversification of investment portfolios, a venture capital firm has hired 20 DMs, including e_1 to e_{20}, to evaluate five potential investment projects: green logistics x_1, ecological materials x_2, digital economy x_3, sustainable energy x_4, and machine learning x_5. The rationale behind involving multiple DMs stems from the recognition that diverse perspectives can significantly enhance the decision-making process, leading to more robust and well-rounded investment strategies. Given that the DMs have different areas of expertise, they are allowed to submit partial IFPRs. In contemporary times, the extensive development of social media makes it easy to expand the number of participants in decision-making to tens of thousands. Nevertheless, considering that the number of DMs in most practical management issues usually does not exceed 100, and the basic principles of this method are not limited by the number of DMs. So, for the sake of simplicity, this paper only uses a case involving 20 DMs for illustration and simulates scenarios involving 50 to 5000 DMs in Fig. 6.

(1) When calculating the degree of immutability, the parameter β controlling the adjustment cost is set to 0.4, which influences the contribution of cost adjustment to the calculation of immutability. Additionally, in Section 6.1.2, we conduct an analysis of β and determine that a value of 0.4 is appropriate.

(2) The division ratio λ between the Bnd and the Neg is set to 0.5, a parameter used to determine the relative importance of the Bnd and Neg in decision-making. It needs to be emphasized that we can dynamically adjust the value of λ based on the urgency of the situation.

(3) The ideal consensus threshold θ_{ideal} is set to 0.92, representing the desired level of consensus in the decision-making process. Due to practical considerations, achieving complete consensus (i.e., 1) is nearly impossible, hence we choose a higher threshold of 0.92 as the most appropriate target.

(4) The parameter K for the KNN algorithm determined through experimental analysis is 4, which decides the number of nearest neighbors considered in the KNN algorithm. Moreover, in Section 6.1.1, we conduct an analysis of K and determine that $K = 4$ is the most suitable value here.

5.2. Solving process

In this section, we will apply the established model to obtain a consistent solution. The specific implementation process includes the following steps:

Step 1: Pre-processing of decision-making information.

In our study, we adopt the standardized evaluation matrix of 20 DMs as utilized by Gong et al. (2024) in their research, along with the social trust relationships among these DMs. The pertinent data and the methodology are meticulously presented in Online Appendix 1 in Gong et al. (2024). Furthermore, we employ a computer program to randomly generate the UAC for these 20 DMs, ensuring a robust and unbiased approach to our analysis.

Step 2: Transforming IFPRs into FPRs.

Step 2.1: After the preference assessment by 20 DMs on 5 alternative options, we collect 20 IFPR matrices and identify the specific locations of 39 missing values and the set M represents the collection of all missing values.

$$\begin{aligned} M = \{\chi \mid & f^1_{12}, f^1_{14}, f^1_{23}, f^2_{12}, f^2_{13}, f^2_{45}, f^3_{15}, f^3_{24}, f^3_{45}, f^4_{13}, f^4_{34}, f^6_{13}, f^6_{15}, \\ & f^6_{45}, f^7_{15}, f^7_{24}, f^7_{34}, f^8_{14}, f^8_{34}, f^9_{13}, f^9_{35}, f^{10}_{24}, f^{10}_{45}, f^{11}_{13}, f^{11}_{23}, f^{11}_{34}, \\ & f^{12}_{25}, f^{13}_{14}, f^{13}_{25}, f^{14}_{24}, f^{14}_{45}, f^{15}_{14}, f^{15}_{25}, f^{16}_{24}, f^{17}_{13}, f^{18}_{12}, f^{18}_{15}, f^{18}_{25}, f^{19}_{14}\}. \end{aligned}$$

Step 2.2: According to Eqs. (14) and (16), each missing value corresponding to the reliability r^{μ}_{ij} can be calculated as:

{3.7309, 3.6484, 3.7339, 3.8063, 3.7104, 3.5741, 3.7249, 3.7604,
3.8335, 3.8562, 3.8520, 3.6958, 3.8595,
3.8291, 3.5735, 3.5842, 3.5156, 3.6567, 3.5920, 3.6315, 3.6389,
3.8246, 3.8683, 3.6934, 3.7078, 3.7077,
3.5639, 3.7334, 3.8448, 3.6241, 3.6267, 3.6546, 3.7539, 3.2492,
3.5670, 3.7476, 3.7432, 3.7403, 3.7539}.

Remark 5.1. To enhance clarity and conciseness in presentation, all numerical values in the examples of this paper are rounded to four decimal places. For access to the complete set of unrounded original numerical data, please refer to the appendix in Gong et al. (2024), where all the exact values are meticulously listed.

The missing value with the greatest reliability is quantified at 3.8683 by Eq. (16) and it is accurately pinpointed to the position f^{10}_{45} within the matrix.

In this study, we have set the parameter for the K-nearest neighbor to 4 from Eq. (11). Accordingly, the four closest neighbors to the DM for the missing value under consideration are identified as e_1, e_8, e_{16}, and e_4.

The membership degree $\widetilde{\mu}_u^a$, which signifies the affiliation of the missing value with its 4-nearest neighbors $\widetilde{\mu}_u^1$, $\widetilde{\mu}_u^2$, $\widetilde{\mu}_u^3$ and $\widetilde{\mu}_u^4$, has been assessed and is represented by the vector [0.2549, 0.2507, 0.2503, 0.2440] by Eq. (14).

By using Eq. (17), we have

$$\begin{aligned} f_{45}^{10} &= \widetilde{\mu}_{10}^1 \times f_{45}^1 + \widetilde{\mu}_{10}^2 \times f_{45}^8 + \widetilde{\mu}_{10}^3 \times f_{45}^{16} + \widetilde{\mu}_{10}^4 \times f_{45}^4 \\ &= 0.2549 \times 0.1056 + 0.2507 \times 0.2703 \\ &\quad + 0.2503 \times 0.8182 + 0.2440 \times 0.4257 \\ &= 0.4033. \end{aligned}$$

Thus, the value of the estimated FPR is 0.4033.

Following the steps demonstrated above, and proceeding in the same manner, complete the estimation of all missing values to obtain all the complete FPR matrices.

Step 3: Clustering process.

Use Algorithm 1 to first obtain the initial cluster centroids. Fig. 2(a) represents the trust relationships among all DMs, the intensity of the colors of the lines connecting DMs directly reflects the level of trust. The deeper the blue, the higher the trust. While Fig. 2(b) represents the initial cluster centroids obtained by Algorithm 1, where larger nodes are cluster centroids. In the process of finding initial cluster centroids, those with the minimum trust distance D from known centroids and the distance D less than the standard deviation are considered to belong to that cluster and are given the same color.

In addition, we use k-means clustering in Algorithm 2 to divide the 20 DMs into 6 subgroups: $C_1 = \{e_3, e_4, e_{10}, e_{12}\}$, $C_2 = \{e_2, e_8, e_{13}\}$, $C_3 = \{e_5, e_7\}$, $C_4 = \{e_9, e_{18}\}$, $C_5 = \{e_1, e_{17}, e_{19}, e_{20}\}$, $C_6 = \{e_6, e_{11}, e_{14}, e_{15}, e_{16}\}$.

Based on this, we calculate the FPR matrix for each subgroup. Furthermore, we can also compute the CDs for these subgroups.

$$F_{(1)} = \begin{pmatrix} 0.5000 & 0.4209 & 0.5053 & 0.2422 & 0.5150 \\ 0.3361 & 0.5000 & 0.6314 & 0.5107 & 0.3098 \\ 0.3319 & 0.2557 & 0.5000 & 0.5838 & 0.5146 \\ 0.4528 & 0.4223 & 0.2797 & 0.5000 & 0.5129 \\ 0.3983 & 0.2765 & 0.2931 & 0.4055 & 0.5000 \end{pmatrix},$$

$$F_{(2)} = \begin{pmatrix} 0.5000 & 0.4629 & 0.4430 & 0.3745 & 0.3908 \\ 0.2970 & 0.5000 & 0.4228 & 0.4670 & 0.5689 \\ 0.3906 & 0.2558 & 0.5000 & 0.2917 & 0.4384 \\ 0.4227 & 0.2400 & 0.5159 & 0.5000 & 0.4200 \\ 0.2761 & 0.3947 & 0.3046 & 0.2988 & 0.5000 \end{pmatrix},$$

$$F_{(3)} = \begin{pmatrix} 0.5000 & 0.5261 & 0.5008 & 0.4184 & 0.3206 \\ 0.2531 & 0.5000 & 0.3997 & 0.4792 & 0.6478 \\ 0.4142 & 0.1445 & 0.5000 & 0.5676 & 0.6177 \\ 0.4242 & 0.4293 & 0.3338 & 0.5000 & 0.4988 \\ 0.3431 & 0.1661 & 0.2695 & 0.2954 & 0.5000 \end{pmatrix},$$

$$F_{(4)} = \begin{pmatrix} 0.5000 & 0.7601 & 0.2651 & 0.6054 & 0.6237 \\ 0.1890 & 0.5000 & 0.2354 & 0.6232 & 0.4341 \\ 0.6652 & 0.2066 & 0.5000 & 0.3680 & 0.4887 \\ 0.3091 & 0.2601 & 0.5282 & 0.5000 & 0.6837 \\ 0.3529 & 0.4554 & 0.5013 & 0.2566 & 0.5000 \end{pmatrix},$$

$$F_{(5)} = \begin{pmatrix} 0.5000 & 0.4874 & 0.4257 & 0.5598 & 0.3638 \\ 0.3189 & 0.5000 & 0.4399 & 0.6847 & 0.5638 \\ 0.4244 & 0.2049 & 0.5000 & 0.6545 & 0.4828 \\ 0.2441 & 0.1797 & 0.1800 & 0.5000 & 0.4549 \\ 0.5532 & 0.3490 & 0.1366 & 0.2459 & 0.5000 \end{pmatrix},$$

$$F_{(6)} = \begin{pmatrix} 0.5000 & 0.5708 & 0.6085 & 0.4620 & 0.6842 \\ 0.2459 & 0.5000 & 0.5088 & 0.4617 & 0.4971 \\ 0.3137 & 0.2615 & 0.5000 & 0.3563 & 0.4445 \\ 0.2550 & 0.4678 & 0.4025 & 0.5000 & 0.5620 \\ 0.2182 & 0.2748 & 0.2910 & 0.4082 & 0.5000 \end{pmatrix}.$$

Based on Eqs. (4) and (7), the CDs for all subgroups are calculated as:

$$CD_{(1)} = 0.8474, CD_{(2)} = 0.8683, CD_{(3)} = 0.8692,$$
$$CD_{(4)} = 0.8814, CD_{(5)} = 0.8405, CD_{(6)} = 0.8601.$$

Step 4: Consensus adjustment mechanism in the first stage

In the first stage, we are committed to fostering consensus among members within each subgroup. Considering that the CD of all subgroups currently falls short of the preset threshold θ_{ideal} of 0.92, this indicates that necessary adjustments must be made to the subgroups through a feedback mechanism to enhance the level of consensus.

Taking $C_1 = \{e_3, e_4, e_{10}, e_{12}\}$ as an example, the initial $CD_{(1)}$ is 0.8474, which is smaller than θ_{ideal}, feedback adjustment is required.

Round 1:

From Eq. (31), we know that the Pos is an empty set. Thus, we first consider $CD_{(1)}$ as the Pos and include the opinions of the DMs e_4 and e_{10} in the Neg. After the first adjustment, we obtain new FPR matrices.

$$F_4^* = \begin{pmatrix} 0.5000 & 0.4409 & 0.5253 & 0.2622 & 0.5350 \\ 0.3561 & 0.5000 & 0.6514 & 0.5307 & 0.3298 \\ 0.3519 & 0.2757 & 0.5000 & 0.6037 & 0.5346 \\ 0.4728 & 0.4422 & 0.2997 & 0.5000 & 0.5329 \\ 0.4183 & 0.2965 & 0.3131 & 0.4255 & 0.5000 \end{pmatrix},$$

$$F_{10}^* = \begin{pmatrix} 0.5000 & 0.4009 & 0.5188 & 0.2622 & 0.4950 \\ 0.3561 & 0.5000 & 0.6114 & 0.4907 & 0.2898 \\ 0.3519 & 0.2757 & 0.5000 & 0.5742 & 0.5346 \\ 0.4328 & 0.4422 & 0.2997 & 0.5000 & 0.4929 \\ 0.4183 & 0.2565 & 0.2731 & 0.3855 & 0.5000 \end{pmatrix}.$$

Since no DMs are assigned to the Pos, according to Eqs. (35) and (36), the trust levels of the DMs in the Neg and Bnd will remain unchanged.

At present, the $CD_{(1)}$ is 0.9084, which still does not meet our set θ_{ideal} of 0.92. Therefore, we conduct a second optimization adjustment.

Round 2:

After the modifications in the previous round, the collective opinions of the subgroup C_1 and the perspectives of DMs in the Neg have undergone changes. These alterations might influence the Pos, leading to adjustments in the CAP and consequently affecting the overall dynamics of the Pos, Neg, and Bnd.

The Pos, Neg and Bnd are re-divided. In this adjustment, we include the opinions of the DMs e_4 and e_{10} in the Pos, while the opinion of the DM e_3 is considered in the Neg. After the optimization process, the DM e_3 has revised his/her opinion as follows:

$$F_3^* = \begin{pmatrix} 0.5000 & 0.4052 & 0.5428 & 0.2580 & 0.4992 \\ 0.3361 & 0.5000 & 0.6157 & 0.4894 & 0.3098 \\ 0.3076 & 0.2314 & 0.5000 & 0.5721 & 0.4903 \\ 0.4528 & 0.4623 & 0.2955 & 0.5000 & 0.5129 \\ 0.4141 & 0.2365 & 0.3173 & 0.4224 & 0.5000 \end{pmatrix}.$$

Updating the trust degree using Eqs. (35) and (36), the trust level of the DM e_3 in the Neg towards the DM e_4 in the Pos is updated as follows:

$$t_{3,4}^* = t_{3,4}^{1-t_{3,4}} = 0.9809^{1-0.9809} = 0.9996.$$

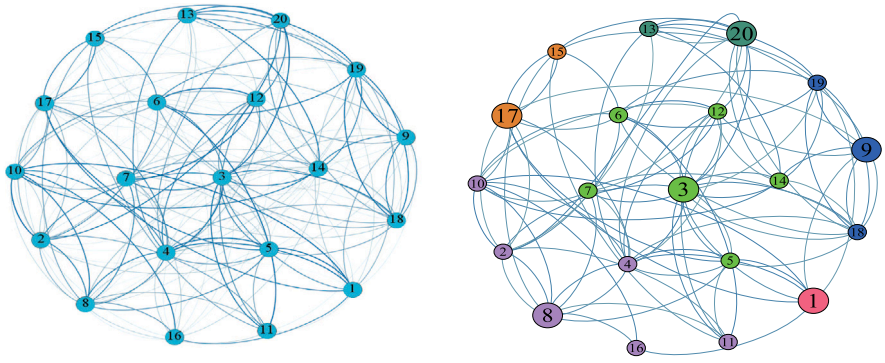
The trust level of the DM e_3 in the Neg towards the DM e_{10} in the Pos is updated:

$$t_{3,10}^* = t_{3,10}^{1-t_{3,10}} = 0.6463^{1-0.6463} = 0.8569.$$

The trust level of the DM e_{12} in the Bnd towards the DM e_3 in the Neg is updated:

$$t_{12,3}^* = t_{12,3}^{1-t_{12,3}} = 0.8491^{1-0.8491} = 0.9756.$$

Through these adjustments, we obtain the updated FPR matrix of C_1 as follows:



(a) Initial STN for 20 DMs (b) Initial cluster centroids obtained by Algorithm 1

Fig. 2. Schematic diagrams of the clustering process.

$$F^*_{(1)} = \begin{pmatrix} 0.5000 & 0.4964 & 0.4584 & 0.2065 & 0.5475 \\ 0.3003 & 0.5000 & 0.6923 & 0.5677 & 0.3415 \\ 0.3683 & 0.2125 & 0.5000 & 0.5928 & 0.5955 \\ 0.4730 & 0.3703 & 0.2434 & 0.5000 & 0.5359 \\ 0.3645 & 0.3182 & 0.2425 & 0.3666 & 0.5000 \end{pmatrix}.$$

The $CD_{(1)}$ has now reached a level of 0.9220, successfully surpassing the predefined θ_{ideal} of 0.92. With this achievement, the consensus formation process for C_1 has been successfully concluded.

Following the same approach as applied to $CD_{(1)}$, we proceed to adjust the remaining subgroups accordingly, ensuring that each one meets the required θ_{ideal}.

Overall, after subgroup clustering is completed, it is observed that some subgroups have not reached the preset consensus threshold. For these subgroups, the approach is to increase the trust among DMs within the subgroup and to guide their opinions towards the collective opinion of the subgroup, as a method for modifying opinions. After one round of adjustment, the consensus level of the subgroups is reassessed. If the consensus level still does not meet the threshold, the process enters a second round of modifications, which will continue until consensus is achieved within each subgroup.

Step 5: Consensus adjustment mechanism in the second phase

Once consensus is reached within all subgroups, the process moves to the second phase, namely the consensus adjustment at the group level. If the global group consensus does not meet the threshold, each subgroup is treated as a single DM, and adjustments are made according to the methods of the first phase, continuing until consensus is achieved across the entire group.

At the beginning of this step, it has already been ensured that consensus within each subgroup has reached the threshold.

Firstly, calculate the trust levels between the various subgroups as:

$$T = \begin{pmatrix} 1.0000 & 0.6612 & 0.6452 & 0.5517 & 0.5083 & 0.5325 \\ 0.5873 & 1.0000 & 0.5272 & 0.4907 & 0.5886 & 0.6160 \\ 0.6381 & 0.6859 & 1.0000 & 0.5975 & 0.5323 & 0.4047 \\ 0.5688 & 0.4749 & 0.7127 & 1.0000 & 0.6536 & 0.3296 \\ 0.5890 & 0.5720 & 0.5161 & 0.6060 & 1.0000 & 0.4883 \\ 0.6288 & 0.4556 & 0.5002 & 0.4759 & 0.4782 & 1.0000 \end{pmatrix}.$$

The group opinion is

$$\overline{F} = \begin{pmatrix} 0.5000 & 0.5613 & 0.4649 & 0.4218 & 0.5069 \\ 0.2712 & 0.5000 & 0.4978 & 0.5455 & 0.5062 \\ 0.4106 & 0.2249 & 0.5000 & 0.4748 & 0.5332 \\ 0.3560 & 0.3440 & 0.3288 & 0.5000 & 0.5279 \\ 0.3257 & 0.3343 & 0.2747 & 0.3291 & 0.5000 \end{pmatrix}.$$

The GCD is 0.9117 which is less than θ_{ideal}, we proceed to the feedback adjustment process.

Round 1:

Initially, each subgroup is considered as a DM. On this basis, the subgroups C_1, C_2 and C_6 are categorized under the Pos, while the subgroup C_3 is assigned to the Neg. The revised opinion of C_3 is:

$$F^*_{(3)} = \begin{pmatrix} 0.5000 & 0.5327 & 0.5178 & 0.3237 & 0.5086 \\ 0.2880 & 0.5000 & 0.5307 & 0.4790 & 0.5148 \\ 0.3878 & 0.2431 & 0.5000 & 0.4293 & 0.5029 \\ 0.4124 & 0.3944 & 0.3602 & 0.5000 & 0.4753 \\ 0.2667 & 0.3160 & 0.2851 & 0.3657 & 0.5000 \end{pmatrix}.$$

As per Eqs. (39) and (40), the trust level of the subgroup C_3 in the Neg towards the subgroup C_1 in the Pos is updated as follows:

$$t^*_{(3),(1)} = t^{1-t_{(3),(1)}}_{(3),(1)} = 0.6380^{1-0.6380} = 0.8499.$$

The trust level of the subgroup C_3 in the Neg towards the subgroup C_2 in the Pos is updated:

$$t^*_{(3),(2)} = t^{1-t_{(3),(2)}}_{(3),(2)} = 0.6859^{1-0.6859} = 0.8883.$$

The trust level of the subgroup C_3 in the Neg towards the subgroup C_6 in the Pos is updated:

$$t^*_{(3),(6)} = t^{1-t_{(3),(6)}}_{(3),(6)} = 0.4047^{1-0.4047} = 0.5836.$$

The trust level of the subgroup C_4 in the Bnd towards the subgroup C_3 in the Neg is updated:

$$t^*_{(4),(3)} = t^{1-t_{(4),(3)}}_{(4),(3)} = 0.7127^{1-0.7127} = 0.9073.$$

The trust level of the subgroup C_5 in the Bnd towards the subgroup C_3 in the Neg is updated:

$$t^*_{(5),(3)} = t^{1-t_{(5),(3)}}_{(5),(3)} = 0.5161^{1-0.5161} = 0.7261.$$

The current consensus value is 0.9188, which is less than 0.92, indicating that consensus has not been reached. It is necessary to re-partition the Pos, Bnd and Neg.

Round 2:

On this basis, the subgroups C_1, C_2, C_3 and C_6 are categorized under the Pos, while the subgroup C_5 is assigned to the Neg. The revised opinion of C_5 is:

$$F^*_{(5)} = \begin{pmatrix} 0.5000 & 0.5045 & 0.4897 & 0.3504 & 0.5140 \\ 0.3173 & 0.5000 & 0.5263 & 0.5080 & 0.5221 \\ 0.3621 & 0.2352 & 0.5000 & 0.4356 & 0.4994 \\ 0.3744 & 0.3691 & 0.3500 & 0.5000 & 0.4730 \\ 0.2951 & 0.3466 & 0.2790 & 0.3579 & 0.5000 \end{pmatrix}.$$

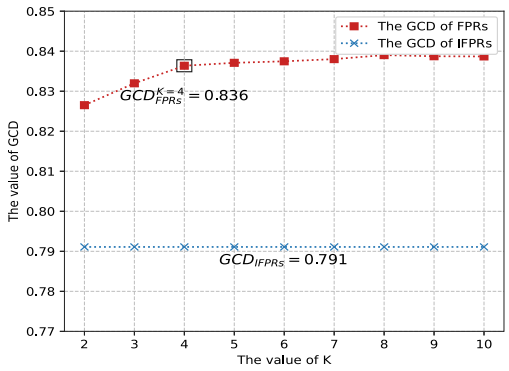


Fig. 3. Comparison among different K values.

The trust level of the subgroup C_5 in the Neg towards the subgroup C_1 in the Pos is updated as follows:

$$t^*_{(5),(1)} = t^{1-t_{(5),(1)}}_{(5),(1)} = 0.5890^{1-0.5890} = 0.8045.$$

The trust level of the subgroup C_5 in the Neg towards the subgroup C_2 in the Pos is updated:

$$t^*_{(5),(2)} = t^{1-t_{(5),(2)}}_{(5),(2)} = 0.5720^{1-0.5720} = 0.7873.$$

The trust level of the subgroup C_5 in the Neg towards the subgroup C_3 in the Pos is updated:

$$t^*_{(5),(3)} = t^{1-t_{(5),(3)}}_{(5),(3)} = 0.7261^{1-0.7261} = 0.9161.$$

The trust level of the subgroup C_5 in the Neg towards the subgroup C_6 in the Pos is updated:

$$t^*_{(5),(6)} = t^{1-t_{(5),(6)}}_{(5),(6)} = 0.4883^{1-0.4883} = 0.6929.$$

The trust level of the subgroup C_4 in the Bnd towards the subgroup C_5 in the Neg is updated:

$$t^*_{(4),(5)} = t^{1-t_{(4),(5)}}_{(4),(5)} = 0.6536^{1-0.6536} = 0.8630.$$

The modified $GCD = 0.9368 > \theta_{ideal} = 0.92$ and the final group opinion is

$$\overline{F} = \begin{pmatrix} 0.5000 & 0.5543 & 0.4896 & 0.3858 & 0.5709 \\ 0.2795 & 0.5000 & 0.5227 & 0.5078 & 0.4950 \\ 0.3863 & 0.2427 & 0.5000 & 0.4152 & 0.5012 \\ 0.3621 & 0.3835 & 0.3746 & 0.5000 & 0.5252 \\ 0.2697 & 0.3397 & 0.3096 & 0.3627 & 0.5000 \end{pmatrix}.$$

The threshold is reached, the second stage feedback adjustment process ends.

Remark 5.2. Experimental results indicate that the algorithm converges in just 1−2 iterations on average, demonstrating high efficiency. The consensus process across both stages completes in a total of 11 rounds, underscoring the algorithm's swift convergence and reliability in practical scenarios.

Step 6: Ranking the solutions.

The scores for each alternative are obtained by Eq. (10), resulting in the following scores for each alternative:

$$Score = [2.5007, 2.3051, 2.0456, 2.1455, 1.7817].$$

Therefore, the final ranking of the alternatives is: $x_1 > x_2 > x_4 > x_3 > x_5$.

6. Comparison and analysis

Within the scope of this section, we undertake an in-depth examination and evaluation of the model's pivotal parameters, simulation outcomes, and comparative analyses, thereby providing a detailed exposition of the strengths and capabilities inherent to our research model.

6.1. Parameter analysis

In this section, we delve into the intricate examination of the parameters that significantly influence the performance of our model. Specifically, we focus on two pivotal parameters: K and β. The analysis aims to elucidate their individual impacts and to determine the optimal settings that enhance the model's efficacy while maintaining computational efficiency. Through a meticulous exploration, we strive to provide a comprehensive understanding of how these parameters interact with the model's mechanisms and contribute to the overall consensus achievement process.

6.1.1. Analysis of the parameter K

In the parameter analysis for missing value estimation, we discover that due to the nature of the KNN algorithm being applied to fill in missing values, there is no dependency on the value of K when no estimation is conducted in Fig. 3. Consequently, the group consensus degree of the IFPR, GCD_{IFPRs}, remains unaffected by variations in K. However, upon initiating the estimation process, the group consensus degree of the FPR, GCD_{FPRs}, notably increases as K increases. We observe that GCD_{FPRs} tends to stabilize when K reaches 4, suggesting that an equilibrium has been achieved. Balancing the enhancement in consensus degree with computational expenses, we determine to set K to 4, ensuring a high consensus degree while also optimizing computational efficiency.

When determining the optimal value of K for cluster analysis, we utilize two evaluation metrics: Dunn's Index (DI) and Davies–Bouldin Index (DBI). In Fig. 4(a), DI tends to have higher values, indicating a higher degree of compactness within clusters. Conversely, in Fig. 4(b), DBI tends to have lower values, signifying better separation between clusters. By calculating these two indices for various values of K, as shown in Fig. 4, we observe that at $K = 4$, DBI reaches a lower value and DI reaches a higher value, demonstrating the best clustering effect. Therefore, we select $K = 4$ as the final clustering parameter to achieve a balance between clustering quality and computational efficiency.

Additionally, it is important to note that the k value for cluster grouping mentioned in Algorithm 2 is adaptively determined, which is a different concept from the K in the KNN algorithm. This distinction is crucial for understanding the adaptive characteristics of the algorithm and the accuracy of the cluster analysis.

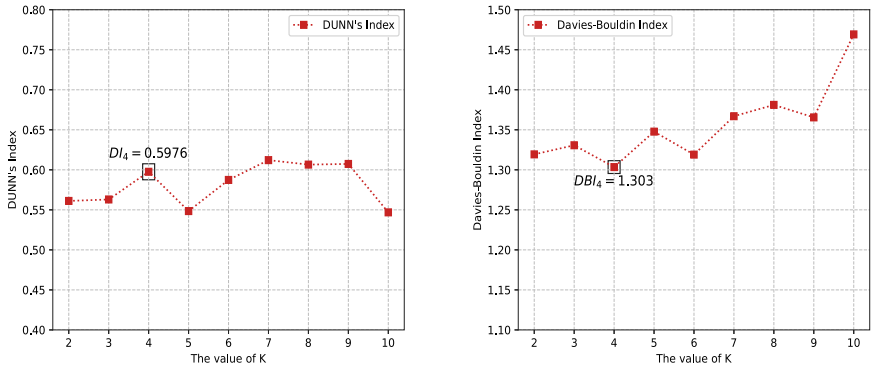
6.1.2. Analysis of the parameter β

The current UAC for the 20 DMs is as follows:

UAC = { 0.37, 0.95, 0.73, 0.6, 0.16, 0.16, 0.06, 0.87, 0.6, 0.71,0.02, 0.97, 0.83, 0.21, 0.18, 0.18, 0.3, 0.52, 0.43, 0.29 }.

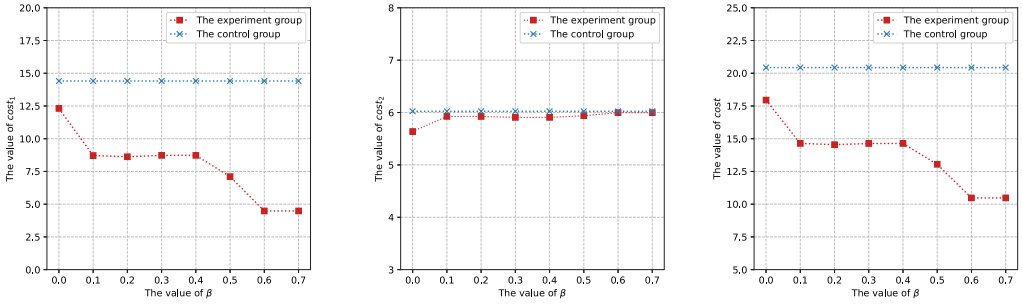
The parameter β plays a decisive role in the division of the Bnd and the Neg when calculating the CAP. To assess the impact of different parameter settings, we establish a control group where the UAC for all DMs are uniformly set to the average value. To ensure the objectivity of the clustering results, we use a simplified formula $d(e_u, e_v) = 1 - MI(e_u, e_v)$ for the distance between elements as per Eq. (27). Similarly, the distances between DMs and subgroups, and between subgroups, are calculated using the formulas $d(e_u, C_l) = 1 - MI(e_u, C_l)$ for Eq. (28) and $d(C_o, C_l) = 1 - MI(C_o, C_l)$ for Eq. (29), respectively.

We name the original experimental setup the "Experimental Group" for easy comparison with the control group. In Fig. 5(a), we present the first-stage cost $cost_1$. In Fig. 5(b), the second-stage cost $cost_2$ and in Fig. 5(c), the total adjustment amount $cost = cost_1 + cost_2$. These



(a) Dunn's Index (b) Davies-Bouldin Index

Fig. 4. Discussion about K.



(a) The first stage cost (b) The second stage cost (c) The total cost

Fig. 5. Discussion about β.

detailed comparative results are comprehensively displayed in Fig. 5, allowing us to fully assess the impacts of different β values on the adjustment of clustering costs. Our experimental results show that when considering the UAC of DMs, the total cost of the Experimental Group in both the first and second stages, as well as the sum of both, is significantly lower than that of the control group. This difference significantly reduces the overall cost, thereby saving resources to a great extent. This finding confirms the effectiveness of our proposed method in cost control and proves the potential value and cost-saving ability of our model in practical applications.

In Fig. 5, we observe that without considering the UAC, when the parameter $\beta = 0$, the cost of the experimental group is comparable to that of the control group, indicating a higher cost burden for the experimental group when the cost parameter is not adjusted. However, as β increases, the cost of the experimental group significantly decreases. This downward trend not only highlights the importance of β as a control parameter in cost optimization but also indirectly confirms the rationality of using the average value for the UAC of the DMs in the control group. Such a setting provides a fair starting point for the experiment, enabling us to assess the actual impact of different β values on cost-effectiveness.

From Fig. 5, it can be seen that as the value of β increases, the adjustment amount of the experimental group decreases, which is consistent with the original intention of setting this parameter. If the UAC is the same, β has a negligible effect on the final adjustment amount. Therefore, considering the importance of trust, preference, and adjustment cost to be roughly equal, this paper selects $\beta = 0.4$ as the parameter value. This choice aims to balance cost-effectiveness while ensuring the flexibility and efficiency of the decision-making process. By setting β to 0.4, our goal is to achieve the best balance between cost control and decision quality, in order to achieve excellent performance in practical applications.

The parameter analysis section serves as a critical foundation for our model's robustness and reliability. By scrutinizing the effects of K and β, we ensure that our model is finely tuned to navigate the complexities of missing value estimation and the calculation of the CAP, thereby fostering a more accurate and efficient consensus formation.

6.2. Simulation experiment

By augmenting the count of DMs and randomly generating pertinent data for our analysis, we have captured the results in a comprehensive and illustrative manner. These findings are elegantly displayed in Figs. 6 and 7, offering a detailed depiction of the effects on our simulation outcomes as the DMs' population expands. This visual representation not only enhances our understanding of the systemic behavior under varying conditions but also provides a robust platform for evaluating the performance and dynamics of the decision-making process within our model.

6.2.1. Consensus simulation analysis

Fig. 6 presents a comparison between the original GCD and the modified GCD after our proposed two-stage CRP. In this process, we expand the number of DMs incrementally by a step size of 50, starting from 50 and scaling up to 5000. The number of alternatives remains constant at 5 in all scenarios. In terms of incomplete information, when there are 50 DMs, there are a total of 10 missing values; with 100 DMs, the number of missing values increases to 20, and so on, with the number of missing values increasing proportionally as the number of DMs grows. Using our method, the final consensus degrees are all able to reach or even exceed the preset θ_{ideal}. This result not only validates the effectiveness of our method but also demonstrates its feasibility and efficiency in handling large groups of DMs. Through this systematic consensus formation mechanism, we can ensure a high degree of consensus unity even when the number of participants increases significantly, thus providing a robust solution for LSGDM.

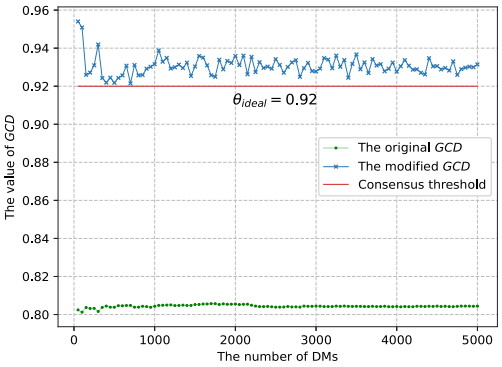


Fig. 6. Comparison of original GCD and modified GCD.

6.2.2. Cost & time simulation analysis

Fig. 7(a) offers a clear visualization, showing that as the number of DMs increases from 50 to 5000, the total cost exhibits a general trend of gradual increase. This reflects the corresponding rise in resource consumption during the decision-making process as the number of participants grows. Fig. 7(b) illustrates the variation in the algorithm's running time as the number of DMs increases. Overall, the running time of the algorithm is satisfactory, demonstrating good time efficiency.

These two figures collectively represent a key part of our simulation experiment, focusing on evaluating the cost-effectiveness and time efficiency of the algorithm when dealing with DM groups of different sizes. Through these detailed data, we can not only gain an in-depth understanding of the algorithm's scalability but also comprehensively assess its performance and feasibility in practical application scenarios. These findings are of significant reference value for guiding further optimization and application deployment of the algorithm.

6.3. Robustness analysis of the missing value estimation method

To assess the robustness and effectiveness of the proposed missing value estimation method, we design and conduct several simulation experiments to adjust the number of DMs p and the dimension of the preference matrix n. Specifically, the number of DMs is set to $p \in \{20, 40, 60, 80, 100\}$, the dimension of preference matrix is set to $n \in \{4, 6, 8, 10, 12\}$, and the probability of missing data is set to three levels of $0.1, 0.2$ and 0.3. Under each combination of p and n, the complete fuzzy preference matrix is first generated and the incomplete matrix is generated by randomly introducing missing values. Subsequently, the method is applied to estimate the missing values to generate the complete preference matrix. Finally, the performance of the method is evaluated using the mean square error (MSE), which is defined as:

$$\mathrm{MSE} = \frac{1}{N}\sum_{j=1}^{N}\left(y_j - \hat{y}_j\right)^2,$$

where y_j denotes the true value, $\hat{y}_j$ denotes the estimated value, and N denotes the total number of missing values.

To ensure the reliability of the results, each experiment is repeated 10 times for each combination of p and n. The estimation performance under different conditions is assessed by calculating the average MSE of the repeated experiments. As shown in Fig. 8, the results show that the estimation error remains within a small range of 0.09 to 0.13 regardless of the number of DMs or the number of scenarios. This consistency validates the robustness of the method in various contexts. In addition, the error hardly changes significantly as the proportion of missing data increases, further demonstrating that the method can still maintain high accuracy when dealing with higher levels of missing data. In summary, the proposed estimation method demonstrates good robustness and stability with low estimation error and the ability to handle missing data effectively.

6.4. Comparison analysis

In this section, we will delve into an in-depth exploration of the intrinsic advantages of the model presented in this study, conducting a comprehensive assessment from both quantitative and qualitative perspectives.

6.4.1. Quantitative analysis

In this paper, we conduct a meticulously designed comparative experiment aimed at demonstrating the superiority of our proposed two-stage consensus reaching mechanism. Fig. 9(a) comprehensively illustrates the consensus reached by all subgroups in this study as well as the overall consensus of the entire system. This result significantly proves that our method not only facilitates the formation of consensus at the system level but also ensures that members within each subgroup can reach a unified opinion. The formation of this two-way consensus not only enhances the stability and reliability of the system but also improves the efficiency and fairness of the decision-making process.

In contrast, the comparative method M_1 proposed by Lu et al. (2022), as shown in Fig. 9(b), while achieving consensus at the group level, failed to achieve this goal at the subgroup level. This indicates that although the method has a certain effectiveness at the macro level, it has obvious limitations at the micro level. Our research results further emphasize the need to consider consensus reaching at both the global and local levels when designing a consensus mechanism to ensure the balanced development and efficient operation of the entire system.

Through this comparative analysis, our method not only demonstrates its innovation and effectiveness in theory but also has significant advantages in practical application. We believe that this two-stage consensus-reaching mechanism will provide strong support for distributed systems, multi-agent systems, and other scenarios that require high levels of collaboration and consensus formation.

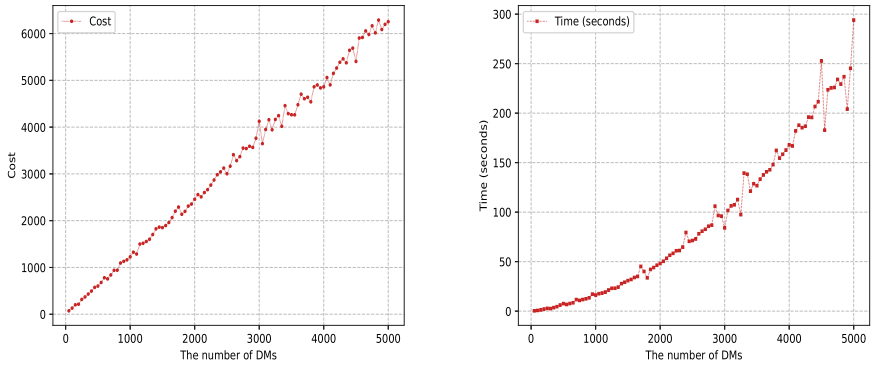
6.4.2. Qualitative analysis

In this part, we review the strengths and unique features of some LSGDM methods that are relevant to this paper and compare with them on a theoretical level, as shown in Table 2.

By observing and analyzing these methods, we draw the following conclusions:

(i) ***In terms of handling incomplete information:*** As the complexity and diversity of information in the decision-making process increases, the emergence of incomplete information becomes more and more common. Such incomplete information may originate from disagreement among DMs, missing information, or ambiguous information. Various techniques have been explored to handle incomplete information, such as methods that take into account opinion similarity (Lu et al., 2022), trust relationships (Chu et al., 2020), and the consistency (Yuan et al., 2023). However, accurately capturing the trust relationships and preference similarities among group members has been critical. For this reason, we design an approach that combines KNN with trust levels and preference similarities. Compared with the existing methods, our constructed method can more accurately reflect the true trust and preference relationships of group members when handling incomplete information, thus enhancing the reliability and validity of information processing.

(ii) ***In terms of clustering distance measurement:*** By integrating multi-factor distance measurement, our method provides a new perspective on handling the mutual influence among DMs. This approach not only takes into account the similarity of preference relationships and the trust degree of DMs but also incorporates the UAC, making the clustering results more comprehensive and accurate. Compared to traditional clustering algorithms that are based solely on a single factor (such as the opinion similarity Shen et al. (2023)), this algorithm is better able to approximate the actual decision-making environment and provide effective solutions for complex decision problems in the real



(a) Cost (b) Time

Fig. 7. The values of cost and running time.

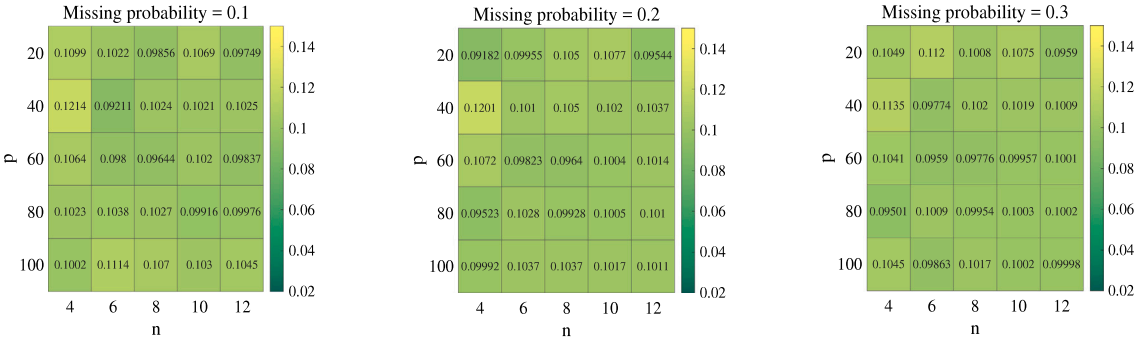
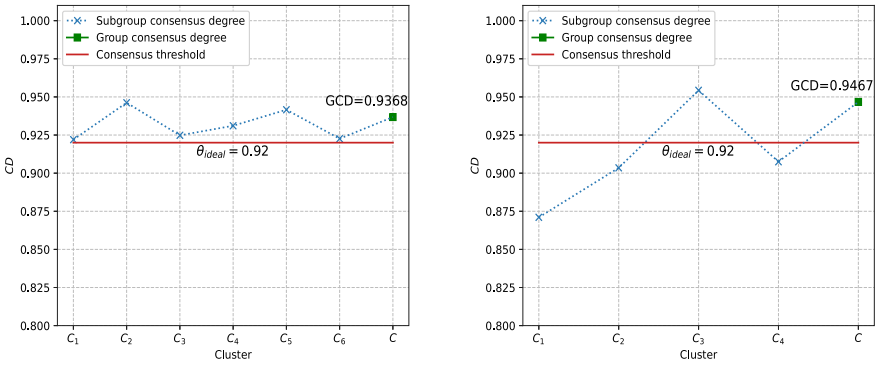


Fig. 8. MSE for different missing probabilities.



(a) The proposed method (b) The comparative method M_1

Fig. 9. Comparison with M_1.

world. Through this multi-dimensional clustering distance measurement, our method can more accurately capture the true relationships between DMs, thereby enhancing the reliability and effectiveness of information processing.

(iii) ***In terms of the consensus mechanism design:*** With the increase in the scale and complexity of decision-making, designing an effective consensus strategy becomes crucial. Currently, various methods adopt different focuses in designing consensus strategies, but realizing efficient and dynamically adjustable consensus strategies has been a challenge (Shen, Ma, Xu et al., 2024; Shen, Ma, Zhang, & Zhan, 2024). In this regard, we design a strategy that combines minimum cost and TWD. Notably, the method is capable of dynamically adjusting the trustworthiness and thus responding more flexibly to the changing opinions and trust relationships in the decision-making process. Compared with the existing methods, our method demonstrates significant advantages in designing consensus strategies that are better adapted to the practical needs of LSGDM.

To sum up, the method presented in this article has an advantage in terms of comprehensiveness of considered factors, covering various aspects such as dynamic trust networks, social networks, minimum adjustment cost, incomplete information, two-stage consensus, and TWD consensus. This suggests that the revealed method of this article may be more comprehensive and effective in handling complex decision-making issues.

7. Conclusions

In the context of big data, this study addresses the consensus achievement issue in SN-LSGDM by proposing a consensus model in light of a dynamic trust network with minimum adjustment cost. The

Table 2
Comparison of different LSGDM methods.

References	Year of publication	Methods of handling incomplete information	Clustering distance measurement	Dynamic social network	Minimum adjustment cost	Consensus strategy
Method 1	2020	Similarity degree and closeness degree	Similarity degree and closeness degree	✓	×	Two-stage iterative modification
Method 2	2022	Trust relationship and opinion similarity	Social distance	×	×	Distrust behaviors identification and management
Method 3	2023	×	Similarity degree	×	✓	Two-stage adaptive minimum cost
Method 4	2023	Reliability and entropy	Decision efficiency distribution	×	×	×
Method 5	2023	Cardinal consistency and ordinal consistency	Similarity degree	×	✓	Minimum cost
Method 6	2024	Reference risk and DMs' weights	Similarity degree and trust degree	✓	×	Dynamic trust and iterative modification
Method 7	2024	Self-confidence degree and DeGroot model	Similarity degree and trust degree	✓	×	Opinion evolution and dynamic trust
Method 8	2024	×	Similarity degree and trust degree	✓	✓	Internal and external consensus
Method 9	2024	×	Trust degree	✓	×	Social network bounded confidence
This paper	–	KNN with trust level and preference similarity	Similarity degree, trust degree and unit adjustment cost	✓	✓	Minimum cost and TWD

Method 1 (Chu et al., 2020).
Method 2 (Lu et al., 2022).
Method 3 (Shen et al., 2023).
Method 4 (Zhou, Wang, Chen, & Zheng, 2023).
Method 5 (Yuan, Wu, & Tu, 2023).
Method 6 (Gong et al., 2024).
Method 7 (Shen, Ma, Xu et al., 2024).
Method 8 (Zhao, Guo, Xu, & Wu, 2024).
Method 9 (Tan, Wang, & Chu, 2024).
"×" describes the factors not mentioned by the method, "✓" describes the factors mentioned by the method.

in-depth analysis and innovative methods of this study provide new perspectives and tools for solving decision-making challenges in the real world. The major contributions of this research are revealed as follows:

(1) This study innovatively introduces a measurement method based on KNN information to fill in missing values. This method comprehensively considers the trust degrees of DMs and the similarity of preference relationships, effectively filling in the incompleteness of preference information, and enhancing the completeness and accuracy of the decision-making process compared to some classical methods.

(2) We adopt an improved k-means clustering algorithm, which not only takes into account the mutual influence among DMs but also the UAC. Compared to some traditional clustering algorithms, it provides a new way to achieve an efficient consensus.

(3) This study designs a two-stage minimum adjustment cost consensus achievement mechanism based on the theory of TWD, using the degree of variability as the standard for division, to achieve feedback adjustment at the individual and subgroup levels, ensuring the coordination and consistency of the decision-making plan.

(4) By introducing an optimization model, we balance the goals of cost minimization and consensus maximization. Through detailed case studies, simulation experiments and comparative analysis, we demonstrate the feasibility and superiority of this method in practical applications. Previous papers mostly did not consider that the unit adjustment amount of each DM might be different, assuming they are the same. Our method is closer to real situations.

In the digital age, LSGDM is crucial for handling big data and complex issues, but it often faces challenges such as incomplete data and divergent opinions. Our research integrates SNs to complete information, employs clustering and consensus adjustment strategies to achieve efficient consensus at minimal cost, and provides an accurate, flexible group decision-making support model. Moreover, our model incorporates dynamic trust networks, considering the adjustment costs of DMs, the similarity of opinions, and trust relationships, to optimize the decision-making process. It is applicable to multiple key fields and offers new solutions for dealing with incomplete information, dynamic trust, and the limitations of clustering algorithms.

Future research will focus on several key areas to further improve the efficiency and quality of consensus achievement below:

Impact of non-cooperative behaviors: We will investigate the manifestations and impacts of non-cooperative behavior within three consensus theories on the decision-making process. By identifying and mitigating potential conflicts, we can better understand the obstacles that may arise during consensus formation and propose corresponding strategies. This research will help us build more robust decision-making models that maintain efficiency and effectiveness in the face of non-cooperative behaviors.

Optimization model adjustments: In response to the current optimization models' inability to reach final consensus in a single iteration, we will re-examine and adjust the objective functions of the optimization models. We will consider introducing new decision factors, such as individual stability, adaptability, and response speed to changes, to enhance the models' adaptability and the likelihood of reaching a consensus. The incorporation of these factors will help the models better adapt to complex and variable decision-making environments, thereby improving the quality and efficiency of decision-making.

Factors beyond trust: Current models rely on trust to align non-consensus decision-makers with the group's opinions. We will explore other factors beyond trust, such as experts' professional knowledge, historical decision-making performance, or social influence, to more

effectively foster alignment between these DMs and the group's opinions. By doing so, we can build more comprehensive decision support systems that consider a variety of factors, not just trust, to achieve a higher quality consensus.

Enhancing model predictive capabilities: We will explore various methods, including machine learning, game theory, or behavioral economics, to enhance the model's predictive capabilities and adaptability to different decision-making scenarios. By integrating theories and techniques from these fields, we can develop more advanced decision support tools that not only predict outcomes but also adapt to the characteristics and preferences of different decision-makers, thereby improving the accuracy and effectiveness of decision-making.

Exploration of cross-domain applications: We will also explore the application of our methodology across different fields to demonstrate its broad applicability and flexibility. This includes applications in industries, business, and policy-making. By applying our methodology in various domains, we can verify its universality and adjust and optimize it according to the characteristics of different fields, further proving the practicality and effectiveness of our methodology.

Through these studies, we expect to further enhance the efficiency and quality of the consensus reaching, providing more scientific and effective support for a variety of decision-making scenarios.

In summary, this study not only enriches the theoretical foundation of LSGDM, but also provides practical tools and methods for consensus achievement in the actual decision-making process. With the continuous development of information technology, we believe that by continuously optimizing and improving the models proposed in this study, we will be able to provide more effective support for solving more complex decision-making problems.

CRediT authorship contribution statement

Jiaxin Zhan: Conceptualization, Methodology, Investigation, Writing – original draft. **Mingjie Cai:** Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

All authors declare that there are no conflicts of interest related to this manuscript. All authors have disclosed any financial and personal relationships that could potentially be viewed as a conflict of interest.

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Data availability

Data will be made available on request.

References

Avramov, D., Cheng, S., & Metzker, L. (2023). Machine learning vs. economic restrictions: Evidence from stock return predictability. *Management Science*, *69*, 2587–2619.

Ben-Arieh, D., & Easton, T. (2007). Multi-criteria group consensus under linear cost opinion elasticity. *Decision Support Systems*, *43*, 713–721.

Ben-Arieh, D., Easton, T., & Evans, B. (2009). Minimum cost consensus with quadratic cost functions. *IEEE Transactions on Systems, Man, and Cybernetics -Part A: Systems and Humans*, *39*, 210–217.

Borlea, I.-D., Precup, R.-E., & Borlea, A.-B. (2022). Improvement of K-means cluster quality by post processing resulted clusters. *Procedia Computer Science*, *199*, 63–70.

Borlea, I.-D., Precup, R.-E., Dragan, F., & Borlea, A.-B. (2017). Centroid update approach to K-means clustering. *Advances in Electrical and Computer Engineering*, *17*(4), 1582–7445.

Chen, H. P., & Xu, G. Q. (2019). Group decision making with incomplete intuitionistic fuzzy preference relations based on additive consistency. *Computers Industrial Engineering*, *135*, 560–567.

Chen, Z. S., Zhu, Z. Z., Wang, Z. J., & Tsang, Y. (2023). Fairness-aware large-scale collective opinion generation paradigm: A case study of evaluating blockchain adoption barriers in medical supply chain. *Information Sciences*, *635*, 257–278.

Chu, J. F., Wang, Y. M., Liu, X. W., & Liu, Y. C. (2020). Social network community analysis based large-scale group decision making approach with incomplete fuzzy preference relations. *Information Fusion*, *60*, 98–120.

Cover, T., & Hart, P. (1967). Nearest neighbor pattern classification. *Institute of Electrical and Electronics Engineers. Transactions on Information Theory*, *13*, 21–27.

Dong, Y. C., Zha, Q. B., Zhang, H. J., Kou, G., Fujita, H., Chiclana, F., et al. (2018). Consensus reaching in social network group decision making: Research paradigms and challenges. *Knowledge-Based Systems*, *162*, 3–13.

Du, Z. J., Luo, H. Y., Lin, X. D., & Yu, S. M. (2020). A trust-similarity analysis-based clustering method for large-scale group decision-making under a social network. *Information Fusion*, *63*, 13–29.

Du, Z. J., Yu, S. M., Luo, H. Y., & Lin, X. D. (2021). Consensus convergence in large-group social network environment: Coordination between trust relationship and opinion similarity. *Knowledge-Based Systems*, *217*, Article 106828.

Du, Z. J., Yu, S. M., & Xu, X. H. (2020). Managing noncooperative behaviors in large-scale group decision-making: Integration of independent and supervised consensus-reaching models. *Information Sciences*, *531*, 119–138.

Garcia-Zamora, D., Labella, A., Ding, W. P., Rodriguez, R. M., & Martinez, L. (2022). Large-scale group decision making: A systematic review and a critical analysis. *IEEE/CAA Journal of Automatica Sinica*, *9*, 949–966.

Gilbert, H. F., & O'Leary, M. H. (1975). Modification of arginine and lysine in proteins with 2 4-pentanedione. *Biochemistry*, *14*, 5194–5199.

Gong, K. X., Ma, W. M., Lei, W. J., & Goh, M. (2024). A dynamic trust network and influence measure-based consensus model for large-scale group decision-making with incomplete intuitionistic fuzzy preference relations. *Journal of the Operational Research Society*, *75*, 1157–1177.

Gou, X. J., Xu, Z. S., & Herrera, F. (2018). Consensus reaching process for large-scale group decision making with double hierarchy hesitant fuzzy linguistic preference relations. *Knowledge-Based Systems*, *157*, 20–33.

Gou, X. J., Xu, Z. S., Liao, H. C., & Herrera, F. (2021). Consensus model handling minority opinions and noncooperative behaviors in large-scale group decision-making under double hierarchy linguistic preference relations. *IEEE Transactions on Cybernetics*, *51*, 283–296.

Herrera-Viedma, E., Cabrerizo, F. J., Kacprzyk, J., & Pedrycz, W. (2014). A review of soft consensus models in a fuzzy environment. *Information Fusion*, *17*, 4–13.

Herrera-Viedma, E., Martinez, L., Mata, F., & Chiclana, F. (2005). A consensus support system model for group decision-making problems with multigranular linguistic preference relations. *IEEE Transactions on Fuzzy Systems*, *13*, 644–658.

Karthik, R., & Ganapathy, S. (2021). A fuzzy recommendation system for predicting the customers interests using sentiment analysis and ontology in e-commerce. *Applied Soft Computing*, *108*, Article 107396.

Ketsarapong, S., Punyangarm, V., Phusavat, K., & Lin, B. (2012). An experience-based system supporting inventory planning: A fuzzy approach. *Expert Systems with Applications*, *39*(12), 6994–7003.

Kilic, U., Sarac Essiz, E., & Kaya Keles, M. (2023). Binary anarchic society optimization for feature selection. *Romanian Journal of Information Science and Technology*, *26*(3–4), 351–364.

Kovalchuk, S. V., Krotov, E., Smirnov, P. A., Nasonov, D. A., & Yakovlev, A. N. (2018). Distributed data-driven platform for urgent decision making in cardiological ambulance control. *Future Generation Computer Systems*, *79*, 144–154.

Li, Y. H., Kou, G., Li, G. X., & Peng, Y. (2022). Consensus reaching process in large-scale group decision making based on bounded confidence and social network. *European Journal of Operational Research*, *303*, 790–802.

Li, M., Qiu, L., Xu, Y., Liu, X., Kou, G., & Herrera-Viedma, E. (2023). A consensus model based on social network analysis in large-scale group decision making: Mining and managing trust risk behaviors. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *53*, 6204–6219.

Liang, D. C., Liu, D., & Kobina, A. (2016). Three-way group decisions with decision-theoretic rough sets. *Information Sciences*, *345*, 46–64.

Liu, Y. J., Liang, C. Y., Chiclana, F., & Wu, J. (2017). A trust induced recommendation mechanism for reaching consensus in group decision making. *Knowledge-Based Systems*, *119*, 221–231.

Liu, X., Xu, Y. J., Montes, R., & Herrera, F. (2019). Social network group decision making: Managing self-confidence-based consensus model with the dynamic importance degree of experts and trust-based feedback mechanism. *Information Sciences*, *505*, 215–232.

Liu, P., Zhang, K., Wang, P., & Wang, F. B. (2022). A clustering- and maximum consensus-based model for social network large-scale group decision making with linguistic distribution. *Information Sciences*, *602*, 269–297.

Long, Q. Q. (2018). Data-driven decision making for supply chain networks with agent-based computational experiment. *Knowledge-Based Systems*, *141*, 55–66.

Lu, Y. L., Xu, Y. J., Herrera-Viedma, E., & Han, Y. F. (2021). Consensus of large-scale group decision making in social network: The minimum cost model based on robust optimization. *Information Sciences*, *547*, 910–930.

Lu, Y. L., Xu, Y. J., Huang, J., Wei, J., & Herrera-Viedma, E. (2022). Social network clustering and consensus-based distrust behaviors management for large-scale group decision-making with incomplete hesitant fuzzy preference relations. *Applied Soft Computing*, *117*, Article 108373.

Ma, W. M., Lei, W. J., & Sun, B. Z. (2020). Three-way group decisions based on multigranulation hesitant fuzzy decision-theoretic rough set over two universes. *Journal of Intelligent & Fuzzy Systems*, *38*, 2165–2179.

Mardani, A., Zavadskas, E. K., Fujita, H., & Koppen, M. (2022). Big data-driven large-scale group decision-making under uncertainty (BiGDM-U). *Applied Intelligence: The International Journal of Artificial Intelligence, Neural Networks, and Complex Problem-Solving Technologies*, *52*, 13341–13344.

Meng, F. Y., Tang, J., & An, Q. X. (2023). Cooperative game based two-stage consensus adjustment mechanism for large-scale group decision making. *Omega*, *117*, Article 102842.

Orlovsky, S. (1978). Decision-making with a fuzzy preference relation. *Fuzzy Sets and Systems*, *1*, 155–167.

Palomares, I., Rodríguez, R. M., & Martínez, L. (2013). An attitude-driven web consensus support system for heterogeneous group decision making. *Expert Systems with Applications*, *40*, 139–149.

Pérez, L. G., Mata, F., & Chiclana, F. (2014). Social network decision making with linguistic trustworthiness-based induced OWA operators. *International Journal of Intelligent Systems*, *29*, 1117–1137.

Protic, D., & Stankovic, M. (2023). XOR-based detector of different decisions on anomalies in the computer network traffic. *Romanian Journal of Information Science and Technology*, *26*(3–4), 323–338.

Shen, Y. F., Ma, X. L., Xu, Z. S., Herrera-Viedma, E., Maresova, P., & Zhan, J. M. (2024). Opinion evolution and dynamic trust-driven consensus model in large-scale group decision-making under incomplete information. *Information Sciences*, *657*, Article 119925.

Shen, Y. F., Ma, X. L., & Zhan, J. M. (2023). A two-stage adaptive consensus reaching model by virtue of three-way clustering for large-scale group decision making. *Information Sciences*, *649*, Article 119658.

Shen, Y. F., Ma, X. L., Zhang, H. J., & Zhan, J. M. (2024). Fusion social network and regret theory for a consensus model with minority opinions in large-scale group decision making. *Information Fusion*, *112*, Article 102548.

Song, Y. M., & Li, G. X. (2019). Handling group decision-making model with incomplete hesitant fuzzy preference relations and its application in medical decision. *Soft Computing*, *23*, 6657–6666.

Sueur, C., Deneubourg, J. L., & Petit, O. (2012). From social network (centralized vs. decentralized) to collective decision-making (unshared vs. shared consensus). *PLOS ONE*, *7*, Article e32566.

Sun, Q., Chiclana, F., Wu, J., Liu, Y. J., Liang, C. Y., & Herrera-Viedma, E. (2023). Weight penalty mechanism for noncooperative behavior in large-scale group decision making with unbalanced linguistic term sets. *IEEE Transactions on Fuzzy Systems*, *31*, 3507–3521.

Tan, J., Wang, Y., & Chu, J. (2024). A consensus method in social network large-scale group decision making with interval information. *Expert Systems with Applications*, *237*, Article 121560.

Tanino, T. (1984). Fuzzy preference orderings in group decision making. *Fuzzy Sets and Systems*, *12*, 117–131.

Wasserman, S., & Faust, K. (1994). *Social network analysis: methods and applications* (1st ed.). Cambridge University Press.

Wu, T., Liu, X. W., Qin, J. D., & Herrera, F. (2021). Balance dynamic clustering analysis and consensus reaching process with consensus evolution networks in large-scale group decision making. *IEEE Transactions on Fuzzy Systems*, *29*, 357–371.

Wu, T., Liu, X. W., Qin, J. D., & Herrera, F. (2022). A new clustering algorithm with preference adjustment cost to reduce the cooperation complexity in large-scale group decision making. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *52*, 5271–5283.

Wu, Q. Q., Tian, D. H., Lan, R. K., & Li, M. (2023). A large-scale MAGDM model based on SKNN and weighted clustering under incomplete information. *Engineering Applications of Artificial Intelligence*, *122*, Article 106158.

Xie, H. T., Ma, Z. M., Xu, Z. S., Fu, Z. W., & Yang, W. (2022). Novel consistency and consensus of generalized intuitionistic fuzzy preference relations with application in group decision making. *Applied Intelligence: The International Journal of Artificial Intelligence, Neural Networks, and Complex Problem-Solving Technologies*, *52*, 16832–16851.

Xu, Y. J., Cabrerizo, F. J., & Herrera-Viedma, E. (2017). A consensus model for hesitant fuzzy preference relations and its application in water allocation management. *Applied Soft Computing*, *58*, 265–284.

Xu, X. H., Du, Z. J., & Chen, X. H. (2015). Consensus model for multi-criteria large-group emergency decision making considering non-cooperative behaviors and minority opinions. *Decision Support Systems*, *79*, 150–160.

Xu, X. H., Du, Z. J., Chen, X. H., & Cai, C. G. (2019). Confidence consensus-based model for large-scale group decision making: A novel approach to managing non-cooperative behaviors. *Information Sciences*, *477*, 410–427.

Xu, Y. J., Li, C. Y., & Wen, X. W. (2018). Missing values estimation and consensus building for incomplete hesitant fuzzy preference relations with multiplicative consistency. *International Journal of Computational Intelligence Systems*, *11*(101).

Xu, Y. J., Wen, X. W., & Zhang, W. C. (2018). A two-stage consensus method for large-scale multi-attribute group decision making with an application to earthquake shelter selection. *Computers & Industrial Engineering*, *116*, 113–129.

Yan, L., Zhao, T., Xie, X. P., & Precup, R.-E. (2024). OSSEFS: An online semi-supervised ensemble fuzzy system for data streams learning with missing values. *Expert Systems with Applications*, *255*, Article 124695.

Yao, Y. Y. (2010). Three-way decisions with probabilistic rough sets. *Information Sciences*, *180*, 341–353.

Yuan, R., Wu, Z. B., & Tu, J. (2023). Large-scale group decision-making with incomplete fuzzy preference relations: The perspective of ordinal consistency. *Fuzzy Sets and Systems*, *454*, 100–124.

Zadeh, L. (1965). Fuzzy sets. *Information and Control*, *8*, 338–353.

Zhang, Z. M., Wang, C., & Tian, X. D. (2016). Corrigendum to multi-criteria group decision making with incomplete hesitant fuzzy preference relations [appl. soft comput. 36 (2015) 1-23]. *Applied Soft Computing*, *40*, 439–440.

Zhang, H. J., Xiao, J., Palomares, I., Liang, H. M., & Dong, Y. C. (2020). Linguistic distribution-based optimization approach for large-scale GDM with comparative linguistic information: An application on the selection of wastewater disinfection technology. *IEEE Transactions on Fuzzy Systems*, *28*, 376–389.

Zhang, H. J., Zhao, S. H., Kou, G., Li, C. C., Dong, Y. C., & Herrera, F. (2020). An overview on feedback mechanisms with minimum adjustment or cost in consensus reaching in group decision making: Research paradigms and challenges. *Information Fusion*, *60*, 65–79.

Zhao, M. K., Guo, J., Xu, Z. S., & Wu, X. H. (2024). A large-scale group decision-making method based on improved affinity propagation algorithm and adjustable minimum-cost consensus model in social networks. *Computers & Industrial Engineering*, *187*, Article 109819.

Zhou, Y. Y., Wang, S., Chen, Y., & Zheng, C. L. (2023). Statistics-based method for large-scale group decision-making with incomplete linguistic distribution fuzzy information: Incorporating reliability and entropy. *Information Fusion*, *99*, Article 101894.